CORE MATHEMATICS GRADE 10: AREA OF POLYGONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.5 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.5: Area of Polygons
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Area of triangles (including the formula ½ × base × height and trigonometric form ½ab sin C) – Area of quadrilaterals: parallelograms, rhombuses, trapeziums and kites – Area of heptagons (7-sided polygons) using decomposition into triangles – Area of octagons (8-sided polygons) using decomposition and formula methods – Area of general irregular polygons using the decomposition/triangulation method – Area of composite/compound polygons (combinations of standard shapes) – Application of area of polygons to real-life situations (land surveying, tiling, farming plots) – Use of geometrical sets, rulers, calculators and triangular cut-outs to measure and verify areas
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) determine the area of triangles using appropriate formulae including the trigonometric form, b) determine the area of quadrilaterals including parallelograms, rhombuses, trapeziums and kites, c) determine the area of heptagons and octagons using decomposition into triangles and standard formulae, d) determine the area of irregular polygons using the triangulation/decomposition method, e) apply knowledge of area of polygons to real-life situations such as land measurement, tiling and farming, f) appreciate the use of area of polygons in solving real-life problems in the Kenyan context.
Core Competencies	– Critical Thinking and Problem Solving: the learner decomposes complex irregular polygons into triangles and standard shapes to determine their areas, and applies these skills to real-life land and tiling problems in Kenyan contexts. – Communication and Collaboration: the learner works in groups to measure, cut out and compare polygon shapes, discusses strategies for decomposing irregular polygons, and presents findings to peers. – Creativity and Imagination: the learner devises multiple methods (decomposition, formula, trigonometric) for finding the area of the same polygon and selects the most efficient approach. – Learning to Learn: the learner connects prior knowledge of triangles and quadrilaterals to build understanding of more complex polygons, and uses rulers, geometrical sets and calculators to verify results independently. – Digital Literacy: the learner uses calculators and available digital devices to compute areas efficiently and to explore polygon area concepts through interactive tools.
Core Values	– Responsibility: the learner carefully handles geometrical sets, rulers and triangular cut-outs, and takes care of shared learning

	resources used to construct and measure polygons. – Integrity: the learner takes accurate measurements and records honest calculations when determining areas of polygons, ensuring results are not fabricated or rounded carelessly. – Respect: the learner appreciates the contributions of every group member during collaborative polygon-decomposition activities and listens to different problem-solving strategies shared by peers. – Unity: the learner works cooperatively in groups to measure and compute areas of polygons representing community spaces such as school plots and farm fields, valuing collective effort.
Science & Engineering Practices	– Using Mathematics and Computational Thinking: the learner selects and applies appropriate area formulae (including trigonometric form) and uses systematic decomposition to compute areas of complex polygons, connecting mathematical reasoning to real-world measurement problems. – Analysing and Interpreting Data: the learner measures polygon dimensions, records data in organised tables, computes areas using multiple methods, and compares results to check for consistency and accuracy. – Constructing Explanations: the learner explains why a polygon can be decomposed into non-overlapping triangles and justifies which formula is most efficient for a given polygon type, communicating reasoning clearly to peers. – Asking Questions and Defining Problems: the learner poses and investigates questions such as "How can I find the area of an irregular shamba boundary?" and defines the mathematical problem before selecting a solution strategy.
Pertinent & Contemporary Issues (PCIs)	– Education for Sustainable Development (Environmental Education): the learner applies area of polygons to calculate the size of farm plots (shamba), forest clearings and community gardens in the local environment, promoting responsible land use and stewardship. – Social Cohesion: the learner works in mixed-ability groups to measure and compute areas of polygons representing shared community spaces, building teamwork and mutual respect across diverse backgrounds. – Financial Literacy: the learner applies area calculations to real-life scenarios such as estimating the cost of tiling a floor in a Nairobi house or fencing a polygonal farm plot in the Rift Valley, connecting mathematics to economic decision-making. – Life Skills (Problem Solving and Decision Making): the learner uses polygon area skills to make informed decisions about land allocation, construction planning and resource estimation in everyday Kenyan contexts.
Career Connections	– Land Surveying and Cartography: surveyors in Kenya use polygon area calculations (including irregular boundary plots) when demarcating land in counties such as Nakuru, Kisumu and the Rift Valley; this sub-strand builds the foundational measurement skills required. – Architecture and Construction: architects and builders apply area of polygons when designing floor plans, tiling patterns and roof structures for homes and public buildings across Kenya. – Agriculture and Farm Management: Kenyan farmers and agricultural extension officers calculate the area of irregularly shaped shamba plots to plan planting, estimate yields of crops like maize and beans, and apply fertiliser accurately. – Graphic Design and Art: designers use polygon area concepts when creating patterns, logos and decorative tile layouts, skills relevant to Kenya's growing creative economy. – Urban Planning and Civil Engineering: planners working on road networks, parks and estates in cities like Nairobi and Mombasa use polygon area methods to allocate land and estimate material quantities.
Focus for Lessons	This unit develops learners' ability to calculate the area of a wide range of polygons — from triangles and standard quadrilaterals through to heptagons, octagons and irregular polygons — using decomposition, formula and trigonometric methods. Anchored in the Kenyan context of land measurement, tiling and urban design, learners move from familiar shapes toward increasingly complex polygonal boundaries, always connecting abstract formulae to tangible real-life situations such as measuring a shamba plot or designing a paving pattern. The pedagogical approach emphasises hands-on measurement with geometrical sets and cut-outs, collaborative group problem solving, and progressive model building in which each lesson's learning adds a new polygon type to a

	growing "area toolkit" that learners apply to an authentic culminating task.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we calculate the area of any polygon — no matter how many sides or how irregular — so that we can fairly divide the Kericho community land plot and solve other real Kenyan land and design problems? KICD KEY INQUIRY QUESTIONS: 1. How do we determine the area of polygons? 2. How is the area of polygons applied in real-life situations?
Anchoring Phenomenon	A community meeting in a village near Kericho is trying to decide how to fairly divide an irregularly shaped piece of land — a polygonal plot with seven unequal sides — among three farming families so that each family receives an equal area for growing tea and maize. The village elder has drawn the boundary on paper, but nobody at the meeting can agree on how to calculate the total area of the plot, let alone divide it equally. The shape does not match any single standard formula the community members remember from school. Students are presented with the drawn polygon boundary (to scale) and asked: "How much land is there in total, and how could you divide it fairly?" This is puzzling because the plot is not a rectangle, triangle or any other single familiar shape — it has seven sides of different lengths and angles — yet the decision about livelihoods and food security depends entirely on getting the area calculation right. The phenomenon motivates learners to investigate whether any polygon, no matter how irregular, can be broken down into shapes whose areas they already know how to find.
Storyline Thread	Lesson 1: ANCHORING PHENOMENON — Students are presented with the Kericho heptagonal land plot. In the Predict Phase, they estimate the area by eye and record individual guesses. The class constructs the Driving Question Board (DQB) listing what they know, what they need to know, and what they wonder. Teacher reviews area of triangles ($\frac{1}{2}bh$) as the foundational "building block" tool and students practise on right-angled and acute triangles using cut-outs, connecting to the idea that any polygon can be "cut up." Lesson 2: QUADRILATERALS — Students observe and measure parallelogram-shaped rooftop and kite-shaped tile cut-outs (supporting phenomena). They derive and apply the area formulae for parallelograms ($A = bh$), rhombuses ($A = \frac{1}{2}d_1d_2$), trapeziums ($A = \frac{1}{2}(a+b)h$) and kites ($A = \frac{1}{2}d_1d_2$) through hands-on measurement activities with rulers and geometrical sets. Groups record results and update the DQB with new tools added to their "area toolkit." Lesson 3: TRIGONOMETRIC AREA OF TRIANGLES — Students investigate the Kisumu rooftop solar panel problem (supporting phenomenon) where the height of a triangle is not directly measurable. They derive and apply the formula $A = \frac{1}{2}ab \sin C$ using known side lengths and an included angle. Calculator and mathematical table skills are practised. Students explain why this formula is needed and update their cumulative polygon-area model. Lesson 4: HEPTAGONS — Students return to the anchoring phenomenon (the seven-sided Kericho land plot). Using triangular cut-outs and rulers, they decompose the heptagon into five non-overlapping triangles, measure each triangle's base and height, compute individual triangle areas and sum them. They compare their computed total with their Lesson 1 estimates and discuss sources of error. DQB is updated: "We can decompose any polygon into triangles!" Lesson 5: OCTAGONS — Students explore the Mombasa octagonal courtyard tiling problem (supporting phenomenon). They decompose a regular octagon into eight isosceles triangles and apply the formula method, then verify with decomposition. They calculate how many square tiles are needed and estimate the cost (integrating financial literacy). Students build an updated cumulative area model diagram showing triangles → quadrilaterals → heptagons → octagons.

	Lesson 6: IRREGULAR POLYGONS — Students tackle a new irregular polygonal plot (5–6 sides, representing a school shamba boundary in Nakuru). They apply the general triangulation method systematically: choose an interior reference point or baseline, draw diagonals, identify triangles, measure and compute. Groups compare answers and reconcile differences. Teacher introduces the coordinate/surveyor's method as an extension for interested learners. Lesson 7: COMPOSITE POLYGONS AND REAL-LIFE APPLICATION — Students work on composite polygon problems: an L-shaped school building footprint and a pentagonal community garden with a rectangular section removed. They combine and subtract standard polygon areas. Groups then return to the anchoring phenomenon and prepare a full, justified solution to the Kericho land-division problem — calculating total area and proposing a fair three-family split with labelled diagrams. Lesson 8: MODEL BUILDING, DQB RESOLUTION AND ASSESSMENT — Groups present their Kericho land-division solutions and defend their decomposition strategies. The class finalises the cumulative area model (a visual "Area of Polygons Toolkit" poster showing all polygon types, formulae and the decomposition principle). The DQB is revisited: all original questions are answered and any remaining wonderings are noted for future learning (e.g., area of circles and sectors in Sub-Strand 2.6). A class written activity assesses individual mastery across all polygon types.
Supporting Phenomena	– A tiler in Mombasa needs to cover an octagonal courtyard floor with square ceramic tiles and wants to know exactly how many tiles to buy — too few means an incomplete job; too many wastes money. How does he calculate the area of the octagonal floor? – The Kenya National Highways Authority (KeNHA) is fencing off a trapezoidal road reserve along the Nairobi–Nakuru highway and needs to know its area to estimate the cost of grass planting. Why can't the engineers simply use length × width? – A school in Kisumu wants to paint a decorative pattern on its wall made up of repeating irregular quadrilateral (kite-shaped) tiles in school colours. The art teacher needs the area of each kite shape to order the correct amount of paint. – A solar energy company installing panels on a parallelogram-shaped rooftop in the Rift Valley needs to calculate the usable roof area accurately to determine how many panels will fit and the expected power output.

LESSON 1 (80 minutes): Anchoring the Phenomenon — The Kericho Land Plot: How Much Land Is There, and Can We Share It Fairly?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon of the irregularly shaped Kericho heptagonal land plot, activate prior knowledge of triangle area as a foundational tool, and open the Driving Question Board so that learners can track their growing understanding across the unit.
Knowledge	Learners will know that: (i) the area of a triangle is calculated using the formula $A = ½bh$; (ii) an irregular polygon does not match a single standard formula; (iii) any polygon can potentially be decomposed into triangles whose individual areas can be summed.
Skills	Learners will be able to: (i) make a reasoned visual estimate of the area of an irregular polygon; (ii) correctly apply $A = ½bh$ to right-angled and acute triangles using measured base and perpendicular height; (iii) articulate what they know, need to know, and wonder about the phenomenon on the DQB.
Attitudes	Learners will appreciate that mathematics is a tool for solving real community problems — including issues of food security, land rights, and fairness — that affect families in Kenya.
Key Inquiry Question	How do we determine the area of polygons?
Purpose in Storyline	This is Lesson 1 of the unit. It anchors the entire storyline by presenting the Kericho heptagonal land plot as an unresolved community problem. Learners discover immediately that their existing knowledge (rectangle, square area formulae) is insufficient, creating a productive knowledge gap. The triangle area review provides the first 'building-block' tool that will later support decomposition of any polygon. The DQB opened today will be revisited and updated in every subsequent lesson.
Safety Notes	Scissors are used for triangle cut-outs. Remind learners to cut away from the body and to pass scissors handle-first. Ensure desks are clear of bags before group activity.

B. LESSON OVERVIEW

Learners are introduced to the anchoring phenomenon: a village community near Kericho must fairly divide an irregularly shaped seven-sided land plot among three farming families growing tea and maize, but nobody at the meeting can calculate the total area because the shape does not match any single standard formula. Students receive a scaled drawing of the heptagonal plot, make individual visual estimates of its total area, and share predictions with the class. The teacher facilitates a whole-class Driving Question Board (DQB) construction in which learners record what they know, what they need to know, and what they wonder. In the second half of the lesson, the teacher reviews the area of a triangle ($A = ½bh$) as the foundational 'building-block' tool, connecting explicitly to the idea that the Kericho plot could eventually be 'cut up' into triangles. Learners practise the formula on right-angled and acute triangle cut-outs, measuring base and perpendicular height, before a closing whole-class synthesis ties every activity back to the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Each learner receives a printed	1. Distribute the scaled plot sheets	Strategy: PREDICTION WITH	OBSERVABLE INDICATOR:

📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	scaled drawing of the Kericho heptagonal land plot (scale 1 cm : 5 m; total plot approximately 1 400 m²). The drawing shows the seven boundary pegs labelled A–G, a north arrow, and a scale bar. Learners study the plot silently for 2 minutes, then write in their exercise books: (a) their individual estimate of the total area in square metres, and (b) one sentence explaining their reasoning strategy. Pairs then compare estimates and discuss: 'Did you use a surrounding rectangle? Did you count grid squares? Did you use a known shape inside the plot?' After pair discussion, the teacher takes a quick show-of-hands poll of estimate ranges (under 800 m², 800–1 200 m², 1 200–1 600 m², over 1 600 m²) and records tallies on the board. A class dot-plot of estimates is sketched, showing the spread and uncertainty. Learners record the class spread in their books alongside their own estimate.	face-down. Say: 'Do NOT turn over until I say go. When you turn it over, you will see a piece of land near Kericho — the same land the families in our story are trying to share. Your job right now is NOT to calculate — just to look and make your best guess.' (Pause 5 seconds.) 'Turn over — GO.' Allow 2 minutes of silent individual thinking. 2. After 2 minutes say: 'Write your estimate and your reasoning in your book. You have 3 minutes. I am not looking for a correct answer — I want YOUR thinking.' Circulate silently; note 3–4 students with contrasting strategies for later cold-calling. 3. Say: 'Now turn to the person next to you. You have 90 seconds — compare your estimates and tell each other HOW you thought about it.' 4. After pair discussion, conduct the show-of-hands poll. As you record tallies say: 'Interesting — we have estimates ranging from [X] to [Y] square metres. That is a difference of [Z] m². For a family whose entire food supply depends on getting an equal share, that uncertainty matters enormously.' 5. Cold-call 3 students with visibly different strategies: 'Amina — walk us through your reasoning.' (WAIT 10 seconds after posing the question before accepting a response.) 'Korir, yours looks different — what did you do?' 'Wanjiku, you got a very different number — can you explain why?' Record the three strategies as labels on the board dot-plot: 'bounding rectangle,' 'grid squares,' 'familiar shape inside.' 6. Close the predict phase: 'None of us can be sure yet. That is exactly the problem the village elder faces. Let us find out what we already	PUBLIC COMMITMENT. By writing and then publicly sharing estimates before any instruction, learners activate prior knowledge of area concepts and commit to a position they will later revise. The class dot-plot makes the range of uncertainty visible, motivating the need for a reliable method. Recording reasoning strategies (bounding rectangle, grid-count, known-shape fitting) surfaces the intuitive decomposition idea the unit will formalise.	Every learner has a written estimate and a reasoning sentence in their book before pair-share begins. During cold-calls, listen for whether learners spontaneously reference familiar shapes (rectangle, triangle) inside the polygon — this reveals their readiness for decomposition. CHECKPOINT QUESTION (ask silently during circulation): 'Can the learner articulate why their method might over- or under-estimate the area?' Flag any learner who cannot write any reasoning for targeted support in Phase 3.

		know — and what we still need to learn.'		
Observe Phase 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 [Search ARES for similar videos] 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	The teacher projects or reads aloud the full phenomenon narrative: a community meeting in a village near Kericho; three farming families (the Chepkemoi family — tea, the Mwangi family — maize, and the Otieno family — mixed crops); the village elder's hand-drawn boundary; the community's disagreement about how to calculate the area. Learners follow along on their scaled drawings. The teacher then displays a short annotated diagram showing the seven boundary measurements (sides of 18 m, 22 m, 15 m, 20 m, 25 m, 17 m, 19 m) and the interior angles, pointing out that no two sides are equal and no angle is 90°. Learners spend 5 minutes in groups of four completing an 'Observation Record' in their books: (i) List every standard area formula you know (rectangle, square, triangle, circle, etc.). (ii) Put a tick next to any formula that could directly give you the area of this plot. (iii) Write one sentence: 'This plot is puzzling because ___.' Groups then do a gallery-walk-style rapid share: each group reads their 'puzzling because' sentence aloud in turn (30 seconds each).	1. Read the phenomenon narrative with expression, pausing at key emotional details: 'The Chepkemoi family has farmed this land for three generations. The Mwangi family moved here after the 2022 drought pushed them out of Baringo. The Otieno family includes a widow raising five children on what she grows. Getting the area wrong does not just mean a calculation error — it means one family goes hungry.' 2. Display the annotated heptagon measurements. Ask: 'Looking at your list of formulae — raise your hand if you found one that works directly for this shape.' (WAIT 12 seconds.) Expected result: no hands, or very few tentative ones. Say: 'That silence tells us something important. We have a real problem that our current toolkit cannot solve — yet.' 3. Circulate during group work. Prompt groups that are stuck: 'Think about the shapes you studied in Grades 7 and 8 — rectangle, triangle, circle. Which of those could you cut OUT of this land plot?' 4. During the gallery walk of 'puzzling because' sentences, cold-call 4 groups. After each, ask the class: 'Does your group agree? Thumbs up, sideways, or down.' Record the most powerful 'puzzling' sentences on a strip of paper — these will go directly onto the DQB in Phase 3. 5. Synthesise: 'Every group has identified the same core puzzle: this plot does not match any single formula we know. So our driving question for this entire unit is —' (reveal the posted driving question banner): 'How can we calculate the	Strategy: PHENOMENON DECONSTRUCTION WITH FORMULA AUDIT. By listing known formulae and auditing which ones apply (none do directly), learners experience the productive knowledge gap firsthand. The emotional context of the three Kenyan farming families makes the mathematical problem feel urgent and real, increasing engagement with the driving question. The 'puzzling because' sentence forces learners to articulate precisely what makes the problem hard — a metacognitive move that primes them for sensemaking.	OBSERVABLE INDICATOR: Each group's Observation Record shows a list of ≥3 formulae, zero or very few ticked as directly applicable, and a 'puzzling because' sentence that references the irregular shape (not just 'it is hard'). LISTEN FOR: any group that ticks 'triangle formula' as applicable — this is a high-readiness group who may already be thinking about decomposition; use them as anchor voices in Phase 3. RED FLAG: any group that cannot list more than one formula — note for pair-support during Phase 3 cut-out activity.

		area of ANY polygon — no matter how many sides or how irregular — so that we can fairly divide the Kericho community land plot and solve other real Kenyan land and design problems?'		
Explain Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — TRIANGLE AREA REVIEW (10 min): Each learner receives two pre-cut card triangles — one right-angled (base 8 cm, height 6 cm) and one acute (base 10 cm, height 5 cm). Learners measure the base and perpendicular height of each using a ruler, then calculate the area using A = ½bh, recording working in their books. They compare answers with their partner. The teacher then demonstrates on the board: for the acute triangle, shows explicitly that the HEIGHT is the perpendicular distance from the base to the opposite vertex — NOT the slant side — and draws the altitude line with a dashed red marker, labelling it h. Learners correct any errors in their own working. PART B — CONNECTING TO THE PHENOMENON (15 min): The teacher draws a simplified version of the Kericho heptagon on the board and says: 'What if we drew lines from ONE vertex — say, peg A — to every other non-adjacent vertex? What shapes would we create inside the plot?' Learners discuss in pairs for 2 minutes, then the teacher draws the diagonals from vertex A, creating five triangles. Learners count: 'A heptagon cut this way gives us FIVE triangles.' Each learner then attempts to identify the base and height of the first triangle (triangle A-B-C) on their scaled drawing, marking them with pencil. Pairs	1. Distribute triangle cut-outs. Say: 'You have 4 minutes — measure base and height of each triangle, calculate the area using A = ½bh, show all working. This is individual work — no sharing yet.' Set a visible timer. 2. After 4 minutes: 'Check with your partner — do your answers match? If they differ, find the error together. You have 90 seconds.' 3. At the board, hold up a large card acute triangle. Ask: 'Baraka — which measurement is the HEIGHT of this triangle?' (WAIT 12 seconds. Cold-call.) After response: 'Show me on the triangle. Come and draw the height line.' Reinforce: 'The height is ALWAYS perpendicular to the base — it makes a 90-degree angle. If you measure the slant side instead, your answer will be wrong — and the Chepkemoi family will get the wrong size plot.' 4. Draw the heptagon on the board. Ask: 'I am going to draw a line from vertex A to vertex C. What shape have I just cut off?' (WAIT 10 seconds.) 'Chiamaka?' After: 'And if I draw A to D?' Continue until all five triangles are visible. Count aloud with class: 'One, two, three, four, five triangles. A seven-sided polygon breaks into five triangles.' 5. Say: 'On your scaled drawing, find triangle A-B-C. Use your ruler to mark the base and the perpendicular height. Work with your partner.' Circulate for 3 minutes; correct any learner who	Strategy: CONCRETE-TO-ABSTRACT BRIDGING WITH DIRECT CONNECTION TO PHENOMENON. Learners physically handle triangle cut-outs (concrete), apply the formula (abstract), and then immediately see the triangle appear inside the Kericho heptagon (connection to phenomenon). The explicit 'building block' sentence at the end of Phase 3 is a sensemaking anchor — a single written claim that learners will test and refine over the coming lessons. Naming the altitude explicitly and distinguishing it from the slant side addresses the most common procedural error before it becomes entrenched.	OBSERVABLE INDICATOR: Both triangle area calculations are correct in learner books with working shown (right-angled: 24 cm²; acute: 25 cm²). The perpendicular height is correctly identified and marked on the acute triangle cut-out. On the scaled heptagon drawing, the base and perpendicular height of triangle A-B-C are correctly marked (height drawn as a dashed line from B perpendicular to line AC, or its extension). MISCONCEPTION CHECKPOINT: If more than 30% of the class marks the slant side as the height on the acute triangle, pause all activity, hold up both a right-angled and an acute triangle side by side, and re-teach the altitude concept using the 'flagpole' analogy: 'The height is like a flagpole standing straight up from the ground — it never leans.'

	check each other's markings. Learners record in their books: 'If I can find the area of each triangle, I can add them up to find the total area of the plot. This means the area of a triangle is the BUILDING BLOCK for finding the area of any polygon.'	marks the slant side as height. 6. Cold-call one pair to explain their markings to the class. 7. Ask the class to complete the 'building block' sentence in their books. Read it aloud together: 'If I can find the area of each triangle, I can add them up to find the total area of the plot.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class constructs the unit's Driving Question Board together on a large sheet of manila paper or a designated section of the classroom wall. The DQB has three columns: WHAT WE KNOW | WHAT WE NEED TO KNOW | WHAT WE WONDER. Each learner receives three sticky notes (or writes three items in their book if physical notes are unavailable). They write: one 'know' item, one 'need to know' item, and one 'wonder' item — all connected to the Kericho land plot phenomenon. Learners post their sticky notes (or the teacher collects and posts representative ones). The class reads through the DQB together, and the teacher organises the 'need to know' items into clusters: (i) How to find area of triangles [already partially answered today], (ii) How to break any polygon into triangles, (iii) How to handle polygons that are not easy to decompose, (iv) How to divide the total area equally among three families. The driving question is posted at the top of the DQB and will remain visible for every lesson in the unit.	1. Introduce the DQB: 'This board is the memory of our class investigation. Every time we learn something new about the Kericho plot, we will update it. Today we start it.' 2. Give thinking time: 'In silence, write your three sticky notes — one KNOW, one NEED TO KNOW, one WONDER. You have 2 minutes.' (WAIT and allow silence.) 3. Invite learners to post. As they post, read each one aloud and ask 'Does everyone agree this belongs in this column? Thumbs up or move?' Correct any that are in the wrong column collaboratively. 4. Cluster the 'need to know' items into 3–4 groups using a coloured marker circle. Name each cluster: 'This group of questions is all about triangle area — we started answering those today. These are all about breaking up polygons — that is the next step. These are about dividing fairly — that will come near the end of our unit.' 5. Point to the driving question banner: 'Every lesson, we will come back to this board and move sticky notes from NEED TO KNOW to KNOW. The day we can answer every question on this board is the day we can finally help the Chepkemoi, Mwangi, and Otieno families.' 6. Cold-call one learner: 'Zawadi — if you had to explain to the village elder what we achieved	Strategy: DRIVING QUESTION BOARD AS A LIVING KNOWLEDGE TRACKER. The DQB externalises metacognition — learners can see what the class collectively knows and does not know. By physically moving sticky notes across the board in future lessons, learners experience knowledge growth as a tangible, visible process. Clustering the 'need to know' questions previews the unit structure and gives learners a sense of agency over the learning trajectory.	OBSERVABLE INDICATOR: Every learner has contributed at least one sticky note (or written entry) in each column. The 'know' entries reference triangle area, polygon sides, or the phenomenon context — not generic statements like 'I know mathematics.' The 'need to know' entries are specific enough to point toward a lesson topic (e.g., 'how do I find the height of a triangle that is not right-angled?' or 'what if I cannot draw a straight line from one vertex to another?'). TEACHER NOTE: Photograph the DQB at the end of this lesson as a baseline. Compare with the DQB at the end of Lesson 6 to document unit-level learning progression for formative records.

		today, what would you say?' (WAIT 12 seconds.) Affirm and extend the response.		
Model Building Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners create their first entry in their 'Unit Model Book' — a dedicated page (or section of their exercise book) that they will revise after each lesson. Today's model is a sketch, not a precise diagram. Each learner draws the Kericho heptagon from memory or using their scaled sheet, and inside it draws the five triangles created by diagonals from vertex A. They label: the overall shape ('7-sided irregular polygon — Kericho land plot'), the five triangles ('T1, T2, T3, T4, T5'), and write the formula A = ½bh next to one triangle. Below the sketch they write: 'My current explanation of how to find the total area: ___.' Most learners will write something like: 'Find the area of each of the five triangles using ½bh, then add them all together.' Learners share their model explanation sentence with a partner and discuss: 'What information would you still need to actually calculate the area of each triangle?' Partners identify: they would need the base and perpendicular height of each of the five triangles. This is recorded as a question on their model page and will be answered in Lesson 2.	1. Say: 'You are now going to draw your first model of how to solve the Kericho land problem. This is not a final answer — it is your best thinking RIGHT NOW. In future lessons you will revise it as you learn more. Open your Unit Model Book to page 1.' 2. Allow 5 minutes of individual sketching and writing. Circulate; prompt any learner who has drawn only the heptagon outline without the interior triangles: 'What did we discover today about breaking up the shape? Can you show that in your drawing?' 3. After 5 minutes, ask learners to share their 'current explanation' sentence with their partner. 4. Whole-class close: cold-call one learner — 'Kipchoge — read us your explanation sentence.' (WAIT 10 seconds.) After reading: 'What would you still need to know to actually use this method on the real Kericho plot?' Collect 2–3 responses. Write on the board: 'In Lesson 2, we will find out how to measure the base and height of each triangle inside any polygon.' 5. Final connection: 'The village elder near Kericho is waiting. Today we have the first piece of the answer — the triangle is our building block. Next lesson, we start building.'	Strategy: CUMULATIVE MODEL REVISION. The Unit Model Book is a running record of each learner's evolving understanding of the phenomenon. Starting with a rough, incomplete model today means learners can see their own intellectual growth lesson by lesson as they add to and revise the model. The 'current explanation' sentence forces learners to translate visual understanding into language — a key sensemaking move. The partner discussion about 'what would you still need?' creates a forward-looking knowledge gap that motivates Lesson 2.	OBSERVABLE INDICATOR: The Unit Model Book entry contains (i) a sketch of the heptagon with ≥3 interior triangles visible, (ii) the formula A = ½bh labelled on at least one triangle, and (iii) a written explanation sentence that mentions both finding triangle areas and adding them. EXIT CHECK: As learners pack away, ask the class (whole-group response): 'Hands up if your explanation sentence mentions ADDING the triangle areas.' If fewer than 70% of hands go up, address the additive step explicitly at the start of Lesson 2 before introducing new content.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your professional journal: (1) PHENOMENON UPTAKE — Did learners connect their mathematical work to the Kericho families' situation, or did the context feel like decoration? Evidence: listen to learner language — are they saying 'the plot' and 'the families' or just 'the shape'? (2) PREDICTION SPREAD — How wide was the range of area estimates? A wide spread (e.g., 400 m² to 2 500 m²) confirms that the phenomenon is genuinely puzzling and that learners cannot rely on intuition alone — this is healthy. A very narrow spread may indicate learners looked at each other's answers. (3) ALTITUDE MISCONCEPTION — How many learners initially marked the slant side as the height of the acute triangle? If more than 30%, plan a 5-minute retrieval activity at the start of Lesson 2 using three contrasting acute triangles before moving to polygon decomposition. (4) DQB QUALITY — Were the 'need to know' questions specific and mathematical, or vague? If vague, model more specific question formation at the start of Lesson 2 by revising 2–3 DQB entries as a class. (5) INCLUSION — Were any learners unable to contribute to the DQB due to language barriers or confidence? Plan a peer-buddy pairing for Lesson 2 to ensure every learner can articulate at least one learning need. (6) PACING — The lesson is 80 minutes; if your school runs 40-minute periods, split at the end of Phase 2 (after the gallery walk) and begin Lesson 1b with a 3-minute recap of the phenomenon before proceeding to Phase 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A scaled drawing of the Kericho seven-sided land plot — an irregular heptagon with seven unequal sides and no right angles. Learners observed that the plot cannot be measured using any single standard formula (rectangle, square, circle) and that the community's inability to calculate the area threatens fair land distribution and food security for three Kenyan farming families.
What did I learn?	The area of a triangle is calculated using $A = ½bh$, where h is the PERPENDICULAR height from the base to the opposite vertex — NOT the slant side. A heptagon can be decomposed into five triangles by drawing diagonals from a single vertex. The total area of the polygon equals the SUM of the areas of all five triangles. This makes the triangle the fundamental 'building block' for finding the area of any polygon.
How does this explain the phenomenon?	If we draw diagonals from one vertex of the Kericho heptagon to all non-adjacent vertices, we divide the plot into five triangles. The total area of the land plot equals the sum of the areas of these five triangles, each calculated using $A = ½bh$. This approach — decomposing any polygon into triangles — is a general strategy that could work for a polygon with any number of sides, which means we have the beginning of a method to help the three farming families near Kericho divide their land fairly.

LESSON 2 (80 minutes): Quadrilaterals — Parallelograms, Rhombuses, Trapeziums and Kites	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To derive and apply the area formulae for parallelograms, rhombuses, trapeziums and kites using hands-on measurement activities, and to add these four quadrilateral tools to the growing 'area toolkit' needed to decompose the irregular seven-sided Kericho community land plot.
Knowledge	Learners will know: (a) the area of a parallelogram is A = bh; (b) the area of a rhombus is $A = \frac{1}{2}d_1d_2$; (c) the area of a trapezium is $A = \frac{1}{2}(a+b)h$; (d) the area of a kite is $A = \frac{1}{2}d_1d_2$; and (e) these formulae are derived by relating each quadrilateral to a rectangle or triangle whose area is already known.
Skills	Learners will be able to: (a) measure bases, heights and diagonals of quadrilateral cut-outs using rulers and geometrical sets; (b) derive each area formula by rearranging or dissecting the shape; (c) calculate areas of parallelograms, rhombuses, trapeziums and kites in contextualised problems; (d) identify which formula applies to a given quadrilateral sub-region of a larger polygon.
Attitudes	Learners will: (a) appreciate that unfamiliar shapes can always be connected to shapes they already understand; (b) show patience and precision in measurement; (c) value collaborative group work as a method of mathematical sense-making; (d) recognise the real-world importance of accurate area calculation for land management and food security in Kenya.
Key Inquiry Question	How do we determine the area of polygons? How is the area of polygons applied in real-life situations?
Purpose in Storyline	In Lesson 1 learners derived the area formulae for triangles and established that any polygon can be decomposed into triangles. In Lesson 2 they extend their toolkit to four quadrilateral shapes — parallelogram, rhombus, trapezium and kite — by measuring rooftop and tile cut-outs. Each new formula is recorded as a new 'tool' on the Driving Question Board, moving the class closer to being able to calculate the total area of the irregular seven-sided Kericho community land plot and divide it fairly among the three farming families.
Safety Notes	Rulers and geometrical set compasses have sharp edges — learners must carry them with the pointed end facing downward and must not wave them around. Scissors used for any cutting activity must be handled blade-down and stored flat on the desk when not in use. Ensure desks are cleared of bags and excess materials to prevent trips.

B. LESSON OVERVIEW

Learners begin by predicting which of four presented quadrilateral cut-outs (a parallelogram shaped like a Kenyan iron-sheet rooftop panel, a rhombus, a trapezium shaped like a tea-farm terrace cross-section, and a kite-shaped decorative tile) has the greatest area. They then carry out a structured hands-on measurement activity in groups of four, using rulers and geometrical sets to measure relevant dimensions of each cut-out and record data in a results table. Through guided dissection (cutting or folding the parallelogram into a rectangle, the trapezium into two triangles, etc.) groups derive each formula, confirm their measurements produce the correct area, and compare with their predictions. In the Explain phase, groups share derivations, the teacher formalises all four formulae, and learners solve two contextualised problems — one involving the floor area of a parallelogram-shaped cattle pen in Naivasha and one involving a trapezium-shaped tea terrace in Kericho. The DQB is updated with four new formula cards, and learners revise their cumulative decomposition model of the seven-sided plot to incorporate the new quadrilateral tools.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Area of a parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner is given a worksheet showing scaled diagrams of four labelled quadrilateral cut-outs: (A) a parallelogram (rooftop panel, base 8 cm, height 5 cm on the diagram), (B) a rhombus (decorative tile, diagonals 10 cm and 6 cm), (C) a trapezium (tea-terrace cross-section, parallel sides 9 cm and 5 cm, height 4 cm), and (D) a kite (festival decoration, diagonals 10 cm and 6 cm). Learners individually write a rank order (largest to smallest area) with a one-sentence justification, then share with a shoulder partner. The class is polled for rank orders — show of hands — and disagreements are noted on the board without resolution. Learners also write one question they have about how to find the area of any of these shapes.	1. Display the four cut-out diagrams on the board (or distribute printed worksheets). Say: 'Before we measure anything, I want you to think carefully — which of these four shapes covers the most area? Write your rank order and ONE reason. You have 3 minutes. Begin.' [Allow 3 minutes of silent individual thinking.] 2. 'Turn to your shoulder partner and compare your ranks. Do you agree? You have 90 seconds.' [Circulate — listen for reasoning that references base×height, diagonal arguments, or purely visual guesses.] 3. Cold-call 4 different learners (one per shape) to state their predicted rank and reason. Say: 'Achieng, what rank did you give shape A and why?' [WAIT 12 seconds after asking before accepting an answer.] Record all four responses on the board as a Prediction Table with columns: Shape | Predicted Rank | Reason. 4. Ask: 'Shapes B and D both have diagonals 10 cm and 6 cm. Raise your hand if you think they have the same area.' [Note the count.] Say: 'Interesting — keep that in mind. We will find out.' 5. Ask learners to write ONE question they still have on a sticky note or the corner of their worksheet. Collect 5–6 by walking around. Read two aloud: 'Here is what some of you are wondering…' Post them on the DQB under 'Questions We Still Have.'	Predict-then-Investigate: By committing to a prediction before gathering evidence, learners activate prior knowledge about area, surface and shape. Disagreements — especially about the rhombus vs. kite — create productive cognitive conflict that motivates the measurement activity. Posting unanswered questions on the DQB models scientific practice of tracking uncertainty.	Observable indicator: Learners can articulate a reason for their ranking that references at least one measurable dimension (base, height, diagonal, or side length) rather than purely visual impression. Teacher listens during pair-share and cold-calls. Red flag: If the majority of reasons are purely 'it looks bigger,' the teacher pauses and asks: 'What measurement would help us decide? What did we learn in Lesson 1 that we could use here?' before proceeding.
Observe Phase	Groups of four receive a set of four card cut-outs (parallelogram,	1. Distribute materials. Say: 'Each group has four cut-outs. Your first	Hands-on Dissection and Measurement (NGSS SEP-3:	Observable indicators: (a) Each group's Dimensions Table is

📹 **VIDEO:** [Area of a parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	rhombus, trapezium, kite) each labelled A–D, plus rulers, a geometrical set (set-square and protractor), pencils, scissors and a results recording sheet. Instructions are on the sheet: (i) For the parallelogram — measure the base and the perpendicular height (use the set-square to locate the true height, not the slant side). (ii) For the rhombus — measure both diagonals. (iii) For the trapezium — measure both parallel sides (a and b) and the perpendicular height. (iv) For the kite — measure both diagonals. Groups record all measurements in the table. They then perform a physical dissection: cut (or fold) the parallelogram along one perpendicular height line and rearrange the two pieces to form a rectangle. They repeat with the trapezium, cutting it into two triangles. They compare the area computed using any trial formula with the area of the reassembled shape. All measurements and observations are recorded. Groups also complete a 'Dimensions Table' that will be used in Phase 3.	job is a careful measurement job — precision matters here, just like it matters for the village elder in Kericho who is measuring the land plot. Sloppy measurements mean unfair land division.' [30 seconds to read the instruction sheet silently.] 2. Circulate continuously. At the 5-minute mark, stop the class: 'Quick check — hold up your set-square. For the parallelogram, where are you placing the set-square to find the height?' [Cold-call one group to demonstrate.] Correct any groups measuring the slant side instead of the perpendicular height. 3. At the 12-minute mark, prompt the dissection step: 'Now for the parallelogram — cut along the dotted line shown. Slide the triangle to the other side. What shape did you make?' [WAIT 10 seconds.] Cold-call: 'Kamau, what shape do you have now, and what are its dimensions?' [WAIT 12 seconds.] 4. At 18 minutes, prompt groups that have finished the parallelogram to move to the trapezium dissection: 'Draw a diagonal on your trapezium to split it into two triangles. Label the triangles 1 and 2. What is the base and height of each triangle?' 5. At 22 minutes, ask all groups to complete their Dimensions Table and do NOT calculate areas yet — save that for Phase 3. Say: 'Your measurements are your evidence. Record them carefully. In Phase 3, you will use them to derive the formulae.'	Planning and Carrying Out Investigations): Physically cutting and rearranging shapes makes the abstract formula derivation concrete. Learners experience that a parallelogram IS a rectangle with the same base and height — they do not just memorise a formula. The use of a set-square to find perpendicular height addresses the most common misconception (confusing slant height with perpendicular height) through direct physical experience rather than teacher correction alone.	completed with numerical values and units (cm). (b) Learners use the set-square correctly to identify perpendicular height on the parallelogram. (c) Learners can name the shape formed when the parallelogram is rearranged. Teacher checkpoints: stop class at 5 min and 12 min as described. Red flag: Groups confusing slant side with height — teacher draws a right-angle symbol and asks: 'The height must make what angle with the base? Use your set-square to check.'
Explain Phase 📹 **VIDEO:** [Area of a parallelogram on	Using their Dimensions Table from Phase 2, groups derive the area formula for each quadrilateral and calculate the area of their specific	1. Begin group formula derivation: 'Using what you saw when you rearranged your shapes, write the formula for each shape's area in	Derive-Before-Apply (NGSS SEP-6: Constructing Explanations): Learners construct the formula from their own dissection evidence	Observable indicators: (a) Each learner's worksheet shows a correct formula for all four quadrilaterals with a brief written

[the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	cut-out. They record: 'The area of a parallelogram = __ because when we rearranged it we got a rectangle of base __ and height __.' Similarly for the rhombus (each diagonal halves the other, giving four right-angle triangles), trapezium (two triangles with bases a and b sharing height h), and kite (same logic as rhombus). Groups then apply their derived formulae to two contextualised problems on the worksheet: (1) A parallelogram-shaped cattle pen near Naivasha has a base of 24 m and a perpendicular height of 15 m. What is its area? (2) A tea-farm terrace in Kericho has a trapezoidal cross-section with parallel widths of 3.6 m and 2.2 m and a depth of 1.4 m. What is the cross-sectional area? Groups write solutions and present findings. The teacher formalises all four formulae on the board with derivation diagrams and connects back to the Kericho land plot.	the box on your worksheet. Use the letters b, h, a, d_1, d_2. You have 6 minutes.' [Circulate — prompt groups who are stuck with: 'What shape did your parallelogram become? What is the area formula for that shape?'] 2. After 6 minutes, cold-call one group per formula (4 cold-calls). For each: 'Group 3, what formula did you derive for the trapezium, and how did you arrive at it?' [WAIT 12 seconds.] Write each derived formula on the board. 3. After all four are on the board, formalise with diagrams: 'Let us confirm. For the parallelogram…' [Draw parallelogram, show the cut, show the rectangle.] Repeat for all four. Say: 'Notice that rhombus and kite share the same formula — $A = \frac{1}{2}d_1d_2$. Why? Raise your hand.' [WAIT 15 seconds.] Cold-call: 'Wanjiku, can you explain why?' [If no correct answer after 15 s, guide: 'Look at the diagonals in both shapes. What do both diagonals do in both shapes?'] 4. Direct learners to the two contextualised problems. Say: 'Problem 1 — a cattle pen in Naivasha. Problem 2 — a tea terrace in Kericho. Solve both individually, then check with your partner. You have 5 minutes.' 5. Cold-call two different learners (one per problem) to present solutions. Require them to state: formula used, substitution, and final answer with units. Correct any unit errors. 6. Connect to phenomenon: 'Look at the Kericho land plot diagram from Lesson 1. Can you now see any sides of the seven-sided plot that, when we draw internal lines, might create a trapezium or	before applying it, which deepens retention and conceptual understanding compared to formula-first teaching. The 'connect back to phenomenon' prompt at the end of Phase 3 ensures sensemaking is always anchored to the driving question. The dual context problems (Naivasha cattle pen + Kericho tea terrace) reinforce that these formulae are tools for real Kenyan land and agricultural problems.	justification referencing the dissection. (b) Problem 1 answer: 360 m². (c) Problem 2 answer: 4.06 m² (accept 4.1 m²). Teacher scans worksheets during circulation. Red flag: If more than one third of learners cannot set up Problem 2, teacher pauses and models the substitution step: 'What are our two parallel sides? What is our height? Let us substitute together.'

		parallelogram? Discuss with your group for 1 minute.' [Take 2 responses.] Say: 'Our toolkit is growing. Let us update the board.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group receives four colour-coded formula cards (pre-printed or hand-written during the lesson): a blue card for the parallelogram formula, a green card for the rhombus, a yellow card for the trapezium, and a red card for the kite. Groups add these four cards to the class DQB in the 'Area Toolkit' column alongside the triangle formula cards from Lesson 1. Each group also writes one new question that today's lesson has raised — for example: 'Can the Kericho plot be divided into only trapeziums and triangles?' or 'What happens if the quadrilateral is irregular — no parallel sides and no equal diagonals?' — and posts it in the 'New Questions' column of the DQB.	1. Point to the DQB. Say: 'In Lesson 1 we added triangle formulas to our toolkit. Today we add four more tools. Come and place your formula cards in the toolkit column.' [Invite one representative per group.] 2. Read the driving question aloud: 'How can we calculate the area of any polygon so that we can fairly divide the Kericho community land plot?' Ask: 'How many tools do we now have in our toolkit? [Count: triangle + parallelogram + rhombus + trapezium + kite = 5.] Is that enough to decompose a seven-sided polygon?' [WAIT 10 seconds.] Accept 2–3 responses. 3. Say: 'Post your new question card. Read the questions from Lesson 1 that are still unanswered. Does today's lesson answer any of them? Give a thumbs-up if yes.' [Briefly review 2 Lesson 1 questions.] 4. Preview Lesson 3: 'Next lesson we look at irregular polygons with more sides — hexagons and heptagons. Will our current toolkit be enough, or do we need new strategies? Keep that question in mind.'	Public Knowledge Construction (DQB Protocol): Making the growing toolkit visible on a shared board externalises the cumulative logic of the unit — learners can literally see their understanding accumulate lesson by lesson. New questions validate intellectual curiosity and model the scientific practice that answering one question always generates new ones (NGSS SEP-1: Asking Questions).	Observable indicator: Each group successfully places all four formula cards in the correct column and the formula written on the card matches the correct algebraic form (e.g., trapezium card reads A = ½(a+b)h, not A = (a+b)h). Teacher verifies each card as it is posted. New question cards are checked for mathematical relevance — vague questions ('What is area?') are redirected: 'Can you make that question more specific to the Kericho plot?'
Model Building Phase 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English	Learners return to the scaled diagram of the seven-sided Kericho community land plot introduced in Lesson 1. In their groups, they attempt to draw internal lines on the plot that decompose it into familiar sub-shapes — now including the four quadrilaterals from today's lesson	1. Distribute clean copies of the Kericho land plot diagram. Say: 'Your job is to divide this seven-sided plot into the shapes in our toolkit. Draw internal lines with a pencil — you can erase and try again. Label each region and write the formula you would use. You have 7 minutes.'	Cumulative Model Revision (NGSS SEP-2: Developing and Using Models): Each lesson, learners physically annotate the same Kericho land plot diagram with new decomposition strategies. This running model makes learning progression visible and tangible. Comparing Group A's 4-	Observable indicators: (a) Each group's plot diagram shows at least one correctly labelled quadrilateral region (parallelogram, rhombus, trapezium or kite) in addition to triangles. (b) The formula written beside each region matches the shape label. (c) Learners can verbally explain why

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	as well as triangles from Lesson 1. Groups aim to produce a decomposition that uses the fewest number of sub-shapes. Each group records their decomposition on a clean copy of the plot diagram, labels each sub-shape (e.g., 'Trapezium 1', 'Triangle 2'), and writes the formula they would use to find each sub-area. Groups do NOT yet calculate numerical areas (since scaled lengths are needed — that comes in a later lesson). They compare decomposition strategies across groups and discuss: 'Is one decomposition better than another?'	2. Circulate actively. Prompt groups that are stuck: 'Look at these two sides here — are they parallel? If yes, what shape might that region be?' [Point without drawing for them.] 3. At 7 minutes, ask two groups with different decompositions to hold up their diagrams. Ask the class: 'Group A used 4 shapes; Group B used 6 shapes. Which decomposition would be easier to work with, and why? Discuss for 1 minute.' 4. Cold-call one learner from each group to justify their choice. [WAIT 12 seconds each time.] 5. Close the lesson: 'Today we added four quadrilateral formulae to our toolkit. We can now decompose most of the Kericho plot. What is still missing from our toolkit that might help? Think about that before Lesson 3.' [WAIT 10 seconds — accept 1–2 responses.] 6. Ask learners to store their annotated plot diagram safely — it will be revised again in Lesson 3.	shape decomposition with Group B's 6-shape version develops mathematical reasoning about efficiency and equivalence — both are correct, but they differ in computational complexity.	they drew an internal line in a particular place. Red flag: Groups that only use triangles (reverting to Lesson 1 toolkit only) — teacher prompts: 'Can you find two sides of the plot that are parallel? That region between them might be a trapezium.'

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners correctly distinguish between slant height and perpendicular height during the measurement activity? If the majority of groups measured the slant side of the parallelogram instead of the true perpendicular height, plan a 5-minute review opener for Lesson 3 using a set-square demonstration on the board. (2) Were learners able to explain WHY the rhombus and kite share the same formula ($A = ½d_1d_2$)? If fewer than half the class could articulate the diagonal-halving reasoning, build in a 3-minute revisit at the start of Lesson 3 using a quick 'convince your partner' task. (3) During the Model Building phase, how many groups successfully incorporated at least one quadrilateral (not just triangles) into their Kericho plot decomposition? If fewer than two groups did so, the Explain phase may not have sufficiently connected the formulae to the phenomenon — consider adding a worked 'spot-the-trapezium-in-the-plot' example at the end of the Explain phase in the next lesson. (4) Did the two contextualised problems (Naivasha cattle pen and Kericho tea terrace) feel authentic and motivating to learners? Note any comments or signs of engagement for adjustment in future lessons. (5) Were group roles (measurer, recorder, presenter, timekeeper) assigned and functioning? If groups were disorganised during the measurement phase, assign explicit roles at the start of Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured the base, perpendicular height, and/or diagonals of four quadrilateral cut-outs (parallelogram shaped like an iron-sheet rooftop panel, rhombus, trapezium shaped like a Kericho tea-farm terrace cross-section, and kite). They physically rearranged a parallelogram into a rectangle and dissected a trapezium into two triangles.
What did I learn?	Learners derived and applied the four area formulae: parallelogram $A = bh$; rhombus $A = ½d_1d_2$; trapezium $A = ½(a+b)h$; kite $A = ½d_1d_2$. They learned that the perpendicular height (not the slant side) must be used in the parallelogram and trapezium formulae, and that the rhombus and kite share a formula because both are fully determined by their two diagonals.
How does this explain the phenomenon?	Learners explained that each quadrilateral formula is derived by linking the shape to a rectangle or pair of triangles whose areas are already known. They applied the formulae to calculate the floor area of a parallelogram-shaped cattle pen in Naivasha and the cross-sectional area of a trapezoidal tea terrace in Kericho, and they updated their decomposition model of the seven-sided Kericho community land plot to include quadrilateral sub-regions alongside triangles.

LESSON 3 (40 minutes): Trigonometric Area of Triangles — When the Height Is Hidden	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students derive and apply the formula A = ½ab sin C by investigating a Kisumu rooftop solar panel problem where the perpendicular height of a triangular panel cannot be directly measured, equipping them with a new tool for finding triangle areas using two sides and their included angle.
Knowledge	– State and derive the formula A = ½ab sin C for the area of a triangle given two sides and the included angle. – Explain why A = ½bh and A = ½ab sin C are equivalent, showing that h = a sin C. – Recognise when the trigonometric area formula must be used instead of A = ½bh (i.e. when the perpendicular height is not directly measurable). – Use mathematical tables and a scientific calculator to evaluate sin of an angle and compute triangle areas.
Skills	– Derive the trigonometric area formula by substituting h = a sin C into the base formula A = ½bh. – Apply A = ½ab sin C to calculate areas of triangles given two sides and the included angle, using both calculators and mathematical tables. – Connect the new formula to the polygon-decomposition strategy developed in Lessons 1 and 2, recognising that any triangle in a decomposed polygon can now be solved even without a measurable height.
Attitudes	– Appreciate that mathematics provides tools to solve real measurement problems that arise in Kenyan engineering and agriculture contexts where direct measurement is impossible. – Show persistence and precision when working with trigonometric values from tables and calculators, understanding that small errors in angle reading produce large errors in area.
Key Inquiry Question	How do we determine the area of polygons — specifically, how can we find the area of a triangle when we cannot measure its perpendicular height directly?
Purpose in Storyline	In Lessons 1 and 2, learners established that any polygon can be decomposed into triangles and that area formulae require perpendicular height. Lesson 3 resolves a critical gap: some triangles inside the Kericho land plot (and in the Kisumu solar panel problem) have no conveniently measurable perpendicular height, but two side lengths and the angle between them ARE known. The trigonometric formula A = ½ab sin C unlocks these 'stuck' triangles, moving learners significantly closer to solving the full seven-sided polygon problem.
Safety Notes	No specific hazards. Ensure calculators are charged and mathematical tables are available for all learners. If working in groups on the floor with large paper diagrams, remind learners to keep the workspace tidy to avoid tripping.

B. LESSON OVERVIEW

This lesson addresses a critical bottleneck in the driving question storyline: learners can decompose any polygon into triangles (Lesson 1), but what happens when the perpendicular height of one of those triangles cannot be measured directly? The lesson is anchored in a vivid supporting phenomenon — a solar energy installation team in Kisumu is fitting triangular solar panels onto a sloped rooftop. The panels are triangular, the workers know two side lengths and the angle between them from their engineering drawings, but they cannot physically measure the perpendicular height because the panel is mounted at an angle against the roof. They need to know the area of each panel to calculate how many panels are needed to power the building and to order the correct amount of materials. This scenario makes it genuinely necessary — not just convenient — to have an alternative area formula.

Learners begin by predicting whether the standard formula A = ½bh is enough to find the area of the solar panel triangle, and most will recognise a problem but not yet have the solution. They then carry out a guided derivation activity: by dropping a perpendicular from one vertex of a labelled triangle and applying the definition of sine from Lesson 2.4 (Trigonometry 1), they discover that the hidden height h equals a sin C, and therefore A = ½bh becomes A = ½b(a sin C) = ½ab sin C. This derivation connects the new lesson directly to prior learning. Learners then practise applying the formula to the Kisumu solar panel problem and to triangles extracted from the Kericho land plot diagram, using both scientific calculators and mathematical tables to find the sine values.

The lesson closes by returning explicitly to the Kericho driving question. Learners update their Driving Question Board and revise their cumulative polygon-area model to include a new tool: the trigonometric area formula. By the end of the lesson, learners can calculate the area of any triangle given two sides and the included angle, removing the last barrier to solving the full decomposed-polygon problem for the Kericho land plot. Teacher reflection questions focus on the quality of the derivation and learner fluency with mathematical tables versus calculators.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a printed card (or teacher draws on board) showing a triangle labelled with two side lengths — 5 m and 7 m — and the included angle of 58°, but NO perpendicular height is marked. The scenario reads: 'A solar panel installer in Kisumu needs to know the area of this triangular panel. The perpendicular height cannot be measured because the panel is bolted flat against a sloped roof. Can you find the area using what you already know?' Learners write their prediction individually for 2 minutes, then share with a shoulder partner. Most will correctly identify that A = ½bh cannot be directly applied because h is unknown; some may guess at h incorrectly; a few may intuit that trigonometry could help.	Display the Kisumu solar panel diagram prominently on the board — label sides a = 5 m, b = 7 m, included angle C = 58°, and write a large question mark where h would be. Say: 'You have 2 minutes — write down whether you think we have enough information to find the area, and what the problem is if we try to use A = ½bh.' [Allow 2 minutes silent individual writing.] After 2 minutes say: 'Turn to your partner and compare predictions. You have 1 minute.' [Allow 1 minute.] Cold-call 3 learners — choose one who appears confident, one who looks uncertain, and one who has written something different from their neighbour. Ask each: 'What is your prediction and what is the key difficulty?' [Allow 10 seconds WAIT TIME after each response before moving on.] Do NOT correct predictions yet. Record the core tension on the board: 'We know TWO SIDES and the ANGLE BETWEEN THEM — but NOT the HEIGHT. Is that enough?' Leave	Predict-then-Share (Individual → Partner → Class): Learners first commit to a written prediction alone, preventing early anchoring to a neighbour's answer. Sharing with a partner surfaces disagreement before whole-class discussion. Cold-calling three contrasting voices (confident, uncertain, different) ensures the class hears the full range of prior knowledge and makes the gap visible to everyone, creating genuine cognitive need for the lesson.	Teacher scans written predictions during the 2-minute individual phase. Key indicators: (1) Does the learner correctly identify that h is the missing quantity? (2) Does the learner link this to A = ½bh from Lesson 1? (3) Does any learner spontaneously mention sine or trigonometry? Learners who cannot articulate the problem with h need targeted support during the derivation phase. Note names of learners who mention trigonometry — they can be strategically cold-called during the Explain Phase to bridge the derivation.

		this as an open question. Connect to storyline: 'This exact problem appears inside the Kericho land plot — some of our decomposed triangles have two known sides and a known included angle, but no measurable height. Today we find the tool to unlock those triangles.'		
Observe Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs on a structured derivation worksheet. The worksheet presents a generic triangle ABC with sides a (opposite A), b (opposite B), c (opposite C), and angle C between sides a and b. Step 1 asks learners to draw the perpendicular height h from vertex A down to side b (extended if necessary) and label it. Step 2 asks: 'Using trigonometry from Lesson 2.4, write an expression for sin C in terms of h and a.' Step 3 asks: 'Rearrange to make h the subject.' Step 4 asks: 'Substitute your expression for h into A = ½bh and simplify.' Step 5 asks: 'Write the final formula.' Learners record every step, then verify by applying their derived formula to the Kisumu solar panel triangle (a = 5 m, b = 7 m, C = 58°) using both a scientific calculator and the mathematical tables on the back of the worksheet. They record both answers and check they match to 2 decimal places.	Distribute the worksheet. Say: 'Work through Steps 1 to 4 with your partner. The only tools you need are what you already know from Trigonometry 1 — specifically the definition of sine in a right-angled triangle, and the formula A = ½bh from Lesson 1. You have 8 minutes.' [Circulate immediately. Do NOT explain the derivation first.] Watch for two specific errors: (A) learners who write sin C = h/b instead of sin C = h/a — these learners have confused which side the height is drawn from; whisper: 'Which side is the height dropped FROM? Label the right-angled triangle you have created.' (B) learners who cannot rearrange to make h the subject — redirect: 'If sin C = h/a, multiply both sides by a. What do you get?' [After 8 minutes, pause whole class.] Cold-call one pair to describe Step 3 aloud. [10 seconds WAIT TIME.] Cold-call a second pair to state the final formula. Write on the board in large text: A = ½ab sin C. Say: 'Now apply this formula to the Kisumu solar panel. Use your calculator first, then the tables. You have 4 minutes.' [Circulate and check calculations.] After 4 minutes, cold-call one pair for their calculator answer and one pair for their table answer. If they differ by more than 0.5 m², pause and model the table reading: 'Find sin	Structured Derivation (Guided Inquiry): Rather than presenting the formula, the worksheet leads learners through the logical steps using only knowledge they already possess (sine definition from Trig 1 + A = ½bh from Lesson 1). This is a 'knowledge-bridging' strategy — the new formula is not received but reconstructed, which greatly improves retention and conceptual understanding. The dual calculation (calculator + tables) builds procedural fluency and confirms that both methods yield the same result, reinforcing trust in the formula.	Circulating teacher checks Step 2 on each pair's worksheet — the key diagnostic is whether sin C = h/a (correct) or sin C = h/b (incorrect side identified). After the whole-class cold-call, teacher checks that the formula written on learners' worksheets matches the board: A = ½ab sin C. For the numerical application, teacher checks: (1) Is sin 58° read correctly from tables (0.8480)? (2) Is the multiplication order correct: ½ × 5 × 7 × sin 58°? (3) Is the final answer approximately 14.84 m²? Pairs whose answers differ between calculator and table by more than 0.5 m² have a table-reading error and should be identified for follow-up.

		58° in the sine column — what row, what column?' Confirm: sin 58° ≈ 0.8480, so A = ½ × 5 × 7 × 0.8480 ≈ 14.84 m².		
Explain Phase 📹 **VIDEO:** [Genetics vocabulary] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Ratios with a Calculator (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners work individually on three application problems, each set in a Kenyan context and each progressively harder. Problem 1 (Direct): A triangular maize plot near Kericho has two sides of 12 m and 9 m with an included angle of 72°. Find its area. Problem 2 (Embedded in polygon decomposition): The same Kericho land plot diagram from Lesson 1 is shown. One of the five decomposed triangles has two marked sides (8.4 m and 11.2 m) and a marked included angle of 65° — but no height. Find its area using A = ½ab sin C. Problem 3 (Challenge — connecting to storyline): If the entire Kericho land plot is to be divided equally among three farming families, and the triangle in Problem 2 is ONE of the five triangles making up the plot, explain why knowing A = ½ab sin C is essential for solving the original driving question. Learners write a full explanation in words for Problem 3.	Say: 'Work through Problems 1 and 2 individually — show every step: write the formula, substitute the values, evaluate using your calculator or tables, and state the answer with units. Problem 3 is a written explanation — not a calculation. You have 10 minutes.' [Allow individual working. Circulate and look specifically at unit usage — many learners drop m² — and at whether learners are correctly identifying which two sides and which angle to use.] After 8 minutes, say: 'Finish your sentence for Problem 3.' [2 minutes.] Cold-call one learner to read their answer to Problem 1 aloud — ask the class: 'Thumbs up if you agree, thumbs sideways if you are unsure.' [10 seconds WAIT TIME.] If any thumbs sideways, cold-call that learner: 'What is different about your answer?' Confirm correct answer: A = ½ × 12 × 9 × sin 72° = ½ × 12 × 9 × 0.9511 ≈ 51.36 m². Cold-call a different learner for Problem 2. Confirm: A = ½ × 8.4 × 11.2 × sin 65° = ½ × 8.4 × 11.2 × 0.9063 ≈ 42.63 m². For Problem 3, cold-call 2 learners to read their written explanations. Prompt with: 'What would happen to the Kericho land division calculation if we could NOT find the area of that triangle?' [10 seconds WAIT TIME.] Reinforce: 'The whole decomposition strategy from Lesson 1 breaks down if even ONE triangle has an unmeasurable height. Today's formula fills that gap.' Write on the	Progressive Contextualisation (Abstract → Concrete → Metacognitive): Problem 1 is a clean calculation to build procedural confidence. Problem 2 embeds the formula directly into the Kericho polygon-decomposition context, forcing learners to see the formula as a tool within the larger storyline. Problem 3 is a metacognitive writing task — learners must articulate WHY the formula matters, which deepens conceptual understanding beyond mechanical application. This three-level structure ensures both procedural fluency and conceptual ownership.	Teacher looks for four quality indicators during circulation: (1) Is the formula written correctly before substitution? (2) Are the correct values substituted for a, b, and C? (3) Is the sine value looked up or calculated correctly? (4) Is the answer given in m² with an appropriate number of decimal places? For Problem 3, teacher assesses whether learners make the explicit connection to the polygon-decomposition strategy — the explanation should mention that not all heights are measurable and that the formula allows them to complete the total area calculation. Collect 5–6 Problem 3 written responses as an exit-ticket sample.

		board: 'KEY INSIGHT: Every triangle in a decomposed polygon CAN now be solved — either with $A = ½bh$ (when h is known) or $A = ½ab \sin C$ (when two sides and the included angle are known).'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Trigonometric Ratios with a Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front of the room. The board already has cards from Lessons 1 and 2 capturing: '$A = ½bh$ for triangles', 'polygon = sum of triangles', 'parallelogram $A = bh$', 'trapezium $A = ½(a+b)h$', 'rhombus/kite $A = ½d_1d_2$'. Learners individually write two new cards: one GREEN card stating what they have LEARNED today (the new formula and when to use it) and one ORANGE card stating a NEW QUESTION this lesson has raised for them about the Kericho land plot or about area in general. They post their cards on the board in 2 minutes. The teacher reads 3–4 orange question cards aloud and briefly comments on which will be addressed in upcoming lessons.	Give each learner one green card and one orange card (or two different-coloured sticky notes / paper strips). Say: 'Green card: in ONE sentence, write the new mathematical tool you gained today and when to use it. Orange card: write ONE new question this lesson has raised — either about the Kericho land plot or about area more generally. You have 2 minutes.' [Allow 2 minutes.] Call two learners to read their green cards aloud. After each, ask: 'Does anyone have a different way of saying the same thing?' [10 seconds WAIT TIME.] Ensure the board now displays: '$A = ½ab \sin C$ — use when two sides and the included angle are known, but height cannot be measured.' Read 3 orange cards aloud. Likely questions will include: 'What if we know the area but not the angle?', 'Can we use this for all five triangles in the Kericho plot?', 'What if we only know one side and some angles?' Respond: 'These questions point us exactly toward Lessons 4 and 5 — the Sine Rule and Cosine Rule. Your questions are driving our sequence forward.' Point to the DQB and say: 'Notice how our board is building a complete toolkit for the Kericho problem — each lesson adds one new tool.'	Driving Question Board as Living Knowledge Record: The DQB functions as a public, cumulative knowledge artefact. Green cards make new learning explicit and shareable; orange cards ensure learner-generated inquiry drives the sequence forward rather than teacher-imposed sequencing. Physically posting cards onto the board creates a visible narrative of growing understanding — learners can see their progress toward the driving question across all three lessons so far.	Teacher reads ALL green cards during the 2-minute posting window (or collects them after the lesson). Key indicator: Does the green card correctly state BOTH the formula ($A = ½ab \sin C$) AND the condition for its use (when height is not directly measurable / two sides and included angle are known)? Cards that only state the formula without the condition reveal surface-level learning. Orange cards are assessed for sophistication — questions that push toward the Sine Rule, Cosine Rule, or general polygon methods indicate strong conceptual progress beyond today's lesson.
Model Building Phase	Each learner has a cumulative 'Polygon Area Model' diagram that began in Lesson 1 as a blank	Distribute learners' cumulative model diagrams (kept in portfolios or exercise books). Say: 'Update	Cumulative Model Annotation (Portfolio as Progressive Artefact): By returning to and revising the	Teacher checks three quality indicators in the updated model: (1) Is the formula correctly written

📹 **VIDEO:** [Genetics vocabulary] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Trigonometric Ratios with a Calculator (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	heptagon and has been progressively annotated. In Lesson 1, they added: 'Decompose into triangles using diagonals from one vertex; use A = ½bh for each triangle.' In Lesson 2, they added quadrilateral formulae as special cases. Today, they add a new annotation to ONE of the five interior triangles in their heptagon diagram: they label two sides and the included angle, cross out the perpendicular height, and write beside it 'Use A = ½ab sin C — height not required.' They also write a one-sentence 'Model Update Statement' at the bottom of their diagram: 'Any triangle in the decomposed polygon can be solved using A = ½ab sin C when two sides and the included angle are known, even if the perpendicular height is inaccessible.' Learners compare their updated model with their partner's to check consistency.	your model in two ways: first, annotate one of the five triangles inside your heptagon with the trigonometric formula situation — label two sides, mark the included angle, and write A = ½ab sin C. Second, write your Model Update Statement at the bottom. You have 3 minutes.' [Circulate and check that learners are annotating the INTERIOR heptagon diagram — not a separate triangle — reinforcing that this formula is a tool for the polygon decomposition.] After 3 minutes, say: 'Show your updated model to your partner. Check: does your partner's annotation clearly show WHICH sides are a and b, and WHICH angle is C? If not, help them label it correctly.' [Allow 1 minute partner check.] Cold-call one learner to hold up their model and explain their annotation to the class in 30 seconds. Ask the class: 'Is their Model Update Statement complete — does it say when to use the formula, not just what the formula is?' [10 seconds WAIT TIME.] Close the lesson by returning to the anchoring phenomenon: 'Look at your Kericho heptagon. Three lessons in, you now have the tools to find the area of any triangle in this plot — whether you can measure the height or not. Next lesson, we will tackle what happens when you know only one side and some angles. The Kericho families are counting on you.'	SAME heptagon diagram across all lessons, learners build a single coherent model of polygon area rather than a set of isolated techniques. Each annotation layer makes the growth of understanding visible. The requirement to annotate an INTERIOR triangle (not a standalone triangle) ensures the new formula is understood as a component of the polygon-decomposition strategy — not as an isolated skill.	as A = ½ab sin C (not A = ½bh)? (2) Is the annotation placed on an interior triangle of the heptagon (showing polygon-level thinking)? (3) Does the Model Update Statement include the condition for use (height not measurable / two sides and included angle known)? Models that only write the formula without the condition or that annotate a separate free-standing triangle (disconnected from the polygon) indicate that the learner has not yet connected today's learning to the overarching storyline. These learners should be prioritised for a 2-minute check-in at the start of Lesson 4.

D. TEACHER REFLECTION

1. During the derivation activity, how many pairs correctly wrote sin C = h/a (not h/b) without prompting — and for those who made the error, was it resolved by the whispered hint or did it persist into the application problems? What does this tell me about learners' fluency with the trigonometric ratio definitions from Lesson 2.4?

2. Were learners able to apply $A = \frac{1}{2}ab \sin C$ to Problem 2 (the triangle embedded inside the Kericho heptagon) with the same confidence as Problem 1 (the standalone maize plot)? If there was a drop in accuracy between Problems 1 and 2, what was the source of the difficulty — the context, the diagram, or the identification of which sides and angle to use? 3. How many learners in the DQB phase wrote green cards that included BOTH the formula AND the condition for its use? What proportion wrote only the formula? What does this gap reveal about the depth of conceptual understanding versus procedural memorisation, and how should I address this at the start of Lesson 4? 4. Did learners demonstrate accurate use of mathematical tables for reading sin values, or did most default to calculators? Given that KNEC assessments require table proficiency, how will I build more structured table-practice time into Lessons 4 and 5 without losing the inquiry pace of the sequence? 5. Looking at the orange DQB cards, did any learners independently raise questions about the Sine Rule, Cosine Rule, or the case where only angles are known — and if so, how can I use those learners' questions as entry points for Lesson 4 rather than a teacher-led introduction?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that a triangle with two known sides (5 m and 7 m) and a known included angle (58°) has a 'hidden' perpendicular height that cannot be measured directly. By drawing the height and applying the sine ratio, they saw that $h = a \sin C$, which when substituted into $A = \frac{1}{2}bh$ produces the formula $A = \frac{1}{2}ab \sin C$. They verified this formula numerically using both a scientific calculator and mathematical tables, confirming that the area of the Kisumu solar panel triangle is approximately 14.84 m^2.
What did I learn?	The formula $A = \frac{1}{2}ab \sin C$ gives the area of any triangle when two side lengths (a and b) and the included angle (C) are known — even when the perpendicular height cannot be measured. This formula is derived directly from $A = \frac{1}{2}bh$ by substituting $h = a \sin C$, so it is not a new rule but an extension of the foundational triangle area formula. A sine value can be found using either a scientific calculator or mathematical tables, and both methods should give the same result.
How does this explain the phenomenon?	The Kericho land plot is decomposed into five triangles. Some of these triangles will have a perpendicular height that is difficult or impossible to measure directly from the boundary drawing — but the side lengths and angles between them CAN be read from the scaled diagram. The formula $A = \frac{1}{2}ab \sin C$ means that EVERY triangle in the decomposed polygon can now be solved, regardless of whether its height is accessible. This removes the last remaining barrier to calculating the total area of the Kericho plot and dividing it fairly among the three farming families.

LESSON 4 (80 minutes): Heptagons — Decomposing the Kericho Land Plot into Triangles	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to determine the area of a heptagon by decomposing it into non-overlapping triangles, connecting directly to the Kericho community land plot.
Knowledge	Learners know that any polygon can be divided into non-overlapping triangles by drawing diagonals from a single interior vertex. The area of each triangle is ½ × base × height, and the total area of the polygon equals the sum of the areas of all constituent triangles.
Skills	Learners accurately measure base lengths and perpendicular heights of triangles using a ruler, calculate individual triangle areas using the formula $A = \frac{1}{2}bh$, sum the areas correctly, and compare their computed total against earlier estimates.
Attitudes	Learners appreciate the power of decomposition as a problem-solving strategy and recognise that accurate measurement and careful calculation have direct consequences for fairness in land allocation and food security in Kenyan communities.
Key Inquiry Question	How do we determine the area of polygons?
Purpose in Storyline	In Lesson 1 learners estimated the area of the seven-sided Kericho land plot. In Lessons 2–3 they mastered area formulae for triangles, quadrilaterals, and pentagons/hexagons. Now in Lesson 4 they return to the anchoring phenomenon — the heptagon — armed with the triangulation strategy, decompose it into exactly five non-overlapping triangles, compute a precise total area, and compare it with their Lesson 1 estimate. This is the pivotal evidence-gathering moment: learners discover that ANY polygon can be broken into triangles, unlocking the general method for solving the fair-land-division problem.
Safety Notes	Rulers have sharp edges — learners should handle them carefully and avoid pointing them at other learners. Scissors (if used to cut triangular templates) must be used pointing downward and stored safely after use.

B. LESSON OVERVIEW
Learners return to the scaled drawing of the seven-sided (heptagon) Kericho community land plot introduced in Lesson 1. Working in groups of four, they use rulers and pre-cut triangular card cut-outs to physically decompose the heptagon into five non-overlapping triangles by drawing diagonals from one chosen interior vertex. They measure each triangle's base and perpendicular height, apply $A = \frac{1}{2}bh$ to find each triangle's area, sum all five areas to obtain the total plot area, convert from cm^2 (map scale) to m^2 (real land), and compare their answer with the estimate they recorded in Lesson 1. Groups then discuss sources of measurement error and update the Driving Question Board with the key insight: "We can decompose ANY polygon into triangles!" The lesson culminates in a class model-building session where learners annotate a shared poster of the heptagon to show the five triangles, their dimensions, and the summed area.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Overview of Ellipses,]	Each learner receives a fresh printed copy of the scaled heptagon (the Kericho land plot boundary, scale 1 cm = 10 m).	Display the heptagon on the board (large poster or projected image). Say: 'This is the same Kericho land plot we first saw in Lesson 1.	Individual written prediction (Predict-Before-Observe). This activates prior knowledge of triangles from Lessons 2–3,	Observable indicator: Every learner has a written prediction with a number (3–7) and at least one sentence of reasoning.

Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Working individually and silently for 3 minutes, learners write a prediction in their exercise books: (a) 'How many non-overlapping triangles do you think you can fit inside this heptagon?' and (b) 'Will your answer today be closer to, or further from, your Lesson 1 estimate? Why?' Learners also circle the vertex on the heptagon from which they plan to draw diagonals.	The village elder is still waiting — the three farming families still cannot agree. Today we are going to get them a precise answer.' Pause 5 seconds. Then say: 'Before you touch a ruler, I want you to predict — silently, individually — how many triangles you think you can cut this shape into, and write one sentence explaining your thinking. You have 3 minutes.' Set a visible timer. After 3 minutes, cold-call 3 learners (select one high-confidence, one mid-level, one quiet learner) to share predictions. Record all three predictions on the board without evaluating them. Say: 'We will come back to these predictions at the end of the lesson.'	surfaces misconceptions about the minimum number of triangles needed, and creates cognitive dissonance that motivates the hands-on evidence activity.	Circulate and note learners who write fewer than 3 or more than 7 — these learners need targeted support in Phase 2. Listen for the misconception that 'you need as many triangles as the number of sides' — address this gently in the whole-class share-out.
Observe Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four receive: one large printed heptagon (scale 1 cm = 10 m), a set of 5 pre-cut blank triangular card cut-outs (labelled T1–T5), a 30 cm ruler, a sharp pencil, and a recording table (columns: Triangle | Base (cm) | Height (cm) | Area (cm²)). Step 1 (5 min): Groups discuss and agree on one interior vertex from which to draw all diagonals. They draw four diagonals, dividing the heptagon into exactly five triangles, and label them T1–T5. Step 2 (10 min): For each triangle, learners place the triangular card cut-out over it to confirm non-overlap and full coverage, then measure the base (any side) and the corresponding perpendicular height with the ruler, recording values to the nearest millimetre. Step 3 (5 min): Each group member independently calculates the area of at least one triangle using A = ½ × base × height,	Circulate continuously. At the 3-minute mark after groups start drawing, pause the class and ask: 'Raise your hand if you chose vertex A as your starting point. Now raise your hand if you chose a different vertex.' Acknowledge variety and say: 'Interesting — let's see if the choice of vertex changes the total area. That itself is a mathematical question worth watching.' At the 8-minute mark, cold-call one group: 'Group 3, tell me — which triangle did you measure first, and what base length did you get?' (Wait 10 seconds after posing the question before accepting the answer.) If any group is drawing overlapping triangles, crouch to their level and say: 'Place your five cut-outs flat on the shape right now. Do any of them overlap? Do they cover every part of the heptagon? Adjust until they do.' For the scale conversion, prompt: 'Your area is in cm². The	Decomposition with Physical Manipulatives (Cut-Out Triangulation). Providing pre-cut triangular card cut-outs makes the abstract concept of polygon decomposition concrete and tactile. The non-overlap/full-coverage check forces learners to confront partitioning rigorously. Recording in a structured table scaffolds the transition from measurement to calculation.	Observable indicators: (1) Each group has exactly five labelled, non-overlapping triangles that tile the heptagon with no gaps. (2) Each row of the recording table has a base, a height, and a correctly computed area (½bh). (3) The five areas sum to a value between 3 500 m² and 6 500 m² (reasonable range for the scaled polygon used). Flag any group whose total is outside this range for immediate one-on-one support. (4) The scale conversion is written explicitly as: 'Area in m² = Area in cm² × 100'.

	recording in cm². Step 4 (5 min): The group collectively sums the five triangle areas to get the total heptagon area in cm², then applies the scale conversion (1 cm = 10 m, so 1 cm² = 100 m²) to find the real plot area in m². Learners record the final answer prominently on their poster sheet.	families need to know their land in m². What does the scale 1 cm = 10 m tell you about areas?' Wait 12 seconds. If no response, say: 'If 1 cm in length equals 10 m in real life, what does 1 cm × 1 cm equal in real life?' Key safety reminder at start of this phase: 'Rulers are tools, not toys — keep them flat on the paper at all times.'		
Explain Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group nominates a 'land surveyor' spokesperson who presents their group's decomposition to the class (2 minutes per group), explaining: how they chose the starting vertex, the five triangle areas, the total, and the converted real-world area. After all groups have shared, learners individually answer three written sensemaking questions in their exercise books: (Q1) 'The minimum number of non-overlapping triangles needed to fill any heptagon is always 5. Why — not 6, not 4?' (Q2) 'Group A got a total of 4 800 m² and Group B got 5 100 m². Both measured the same polygon. What could explain the difference?' (Q3) 'One farming family should receive one-third of the total. Using your group's answer, calculate the area each family should receive.' A class discussion then surfaces the generalisation: for any n-gon, decomposition from one vertex gives (n – 2) triangles.	Before group presentations, say: 'You are presenting to the village elder and the three families. Be precise — lives and livelihoods depend on your number.' During presentations, use the following cold-call protocol for Q1: 'I am going to give you 10 seconds of thinking time. Then I will choose someone to explain why 5 triangles and not 4.' Count silently to 10, then cold-call a mid-level learner. After their response, cold-call a second learner: 'Do you agree? Add something or challenge it.' For Q2 (sources of error), after 12 seconds of wait time ask: 'What are three possible sources of measurement error when using a ruler on a printed polygon?' Accept: parallax error, ruler not aligned to perpendicular height, printed scale distortion. Record these on the board under the heading 'Why our answers differ.' For the generalisation, write on the board: heptagon → 5 triangles; hexagon → 4 triangles (Lesson 3); pentagon → 3 triangles (Lesson 3). Ask: 'What pattern do you see? Write it as a formula.' Wait 15 seconds. Cold-call 3 learners. Guide toward: number of triangles = n – 2. Say: 'This is the key insight. Write it and box it in your notes.'	Pattern Generalisation via Data from Multiple Lessons (Cross-Lesson Evidence Comparison). By deliberately tabulating results from Lessons 2–4 on the board, the teacher helps learners inductively derive the (n – 2) rule. The three written sensemaking questions scaffold progression from recall (Q1) to error analysis (Q2) to application (Q3), aligned with NGSS Science and Engineering Practice 6 (Constructing Explanations) and Practice 4 (Analysing and Interpreting Data).	Observable indicators: (1) Written answer to Q1 includes the phrase or equivalent of 'n – 2 triangles for an n-gon' or a correct numerical justification for 7 – 2 = 5. (2) Written answer to Q2 names at least two legitimate sources of measurement error. (3) Written answer to Q3 shows correct division of the group's total area by 3 with appropriate units (m²). Exit check: teacher collects Q3 answers from three random learners before moving to Phase 4.

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group receives two sticky notes. On the GREEN sticky note they write one thing they now know that they did not know in Lesson 1 (a 'We CAN now...' statement). On the ORANGE sticky note they write one new question that has emerged (a 'We still wonder...' statement). Groups post both sticky notes on the class DQB under the columns 'NEW EVIDENCE' and 'NEW QUESTIONS'. The teacher then reads aloud three green notes and three orange notes, and the class votes (thumbs up) for the green statement that best captures today's big idea. The winning statement — expected to be a variation of 'We can decompose any polygon into triangles!' — is written in large letters on a dedicated DQB card and pinned at the centre of the board. Learners also annotate their personal DQB worksheet (running record from Lesson 1) to record: today's lesson number, the key finding, and the updated best estimate of the plot area.	Distribute sticky notes. Say: 'Green note: finish the sentence — We CAN now... Orange note: finish the sentence — We still wonder... You have 2 minutes.' After posting, read three green notes aloud using a dramatic voice as if addressing the village elder: 'The elder hears from Group 1: [read note].' After the vote, hold up the winning card and say: 'This is our DQB headline for today. Does everyone agree this is what we proved?' Wait 8 seconds for any dissent. Then say: 'Notice how far we have come from Lesson 1 — then we could only estimate. Now we can calculate. What changed?' Cold-call 2 learners. Finally, point to remaining orange sticky notes and say: 'These are our questions for Lessons 5 and 6. We are not done yet — the plot is irregular, and next we will deal with truly irregular polygons.'	Driving Question Board — Cumulative Evidence Tracking. The DQB functions as a public, shared record of the class's growing understanding. Green/orange differentiation trains learners to distinguish settled knowledge from open questions, a core scientific practice (NGSS Practice 1: Asking Questions). Linking today's DQB update explicitly to Lesson 1 estimates reinforces the storyline arc.	Observable indicators: (1) Every group has posted both a green and an orange sticky note with complete sentences. (2) Green notes reference either the triangulation method or the $(n - 2)$ rule — not vague statements like 'we learned about heptagons.' (3) Orange notes reveal productive next-step thinking (e.g., 'What if we cannot draw a perpendicular height easily?' or 'What about a polygon with a dent — a concave shape?'). Teacher photographs the DQB for the digital lesson record.
Model Building Phase 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)]	Groups receive a large A2 poster sheet pre-printed with: (i) the heptagon outline (same scale as all previous lessons), (ii) a blank 'Running Area Model' table with rows for Lessons 1–8 and columns for 'Shape', 'Method Used', 'Area Formula/Strategy', and 'Total Area (m^2)'. Learners: (a) Draw and colour the five triangles on the heptagon, labelling each triangle's base, height, and area. (b) Write the general statement: 'Area of heptagon = A(T1) + A(T2) + A(T3) + A(T4) + A(T5) = $\frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 + \frac{1}{2}b_3h_3 + \frac{1}{2}b_4h_4 + \frac{1}{2}b_5h_5$.' (c) Fill in the Lesson 4 row of the Running	Distribute A2 poster sheets. Say: 'This poster is your team's evidence document. It must be good enough to show to the real village elder — clear, accurate, and labelled.' Circulate and check that (a) triangles are non-overlapping and fully cover the heptagon, (b) base-height pairs are correctly identified (the height must be perpendicular to the chosen base), and (c) the general formula is written symbolically, not just numerically. For the octagon extension sketch, say: 'You have 3 minutes. Quickly sketch an octagon and show me — without	Cumulative Model Revision (Running Area Model Poster). Maintaining a running poster across all 8 lessons is a concrete instantiation of NGSS Practice 2 (Developing and Using Models). Each lesson adds one row to the table and one annotated shape to the poster, so by Lesson 8 learners possess a complete visual and symbolic model of how to find the area of any polygon. The forward-bridge octagon sketch activates anticipatory thinking and makes Lesson 5 feel necessary and motivated.	Observable indicators: (1) The heptagon on the poster shows exactly five coloured, labelled, non-overlapping triangles. (2) The symbolic formula $A = \frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 + ... + \frac{1}{2}b_5h_5$ is written correctly. (3) The Lesson 4 row in the Running Area Model table is fully completed with the group's computed area in m^2. (4) The octagon extension sketch shows 6 triangles (or an attempt at 6). Teacher stamps or initials completed posters before learners store them.

Source: CK-12 🔍 Search ARES for similar readings	Area Model table. (d) Sketch a thought-bubble beside the heptagon showing how the same logic (triangulation) could apply to an octagon — drawing (n – 2) = 6 triangles on a rough octagon sketch. This last step is a deliberate forward-bridge to Lesson 5.	measuring — how many triangles you would need. Draw them in.' After 3 minutes, cold-call one group to hold up their octagon sketch. Ask the class: 'Six triangles — is the class convinced? Someone convince me using the (n – 2) rule.' Wait 10 seconds. Cold-call a different learner than those cold-called in Phase 3. Conclude: 'Next lesson we investigate octagons with measurements. The Kericho families are getting closer to their answer.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did all groups successfully decompose the heptagon into exactly five non-overlapping triangles, or did some groups produce gaps or overlaps? If overlaps occurred, was it because learners chose a reflex-angle vertex — and how will you address this in Lesson 5 when dealing with irregular/concave polygons? (2) How close were the groups' computed areas to each other? If the spread was greater than 15%, was this due to measurement error, scale-reading error, or incorrect identification of the perpendicular height? Plan a 5-minute targeted drill on 'finding the perpendicular height of a triangle' at the start of Lesson 5 if needed. (3) Did the (n – 2) generalisation emerge naturally from the class, or did you have to supply it? If you supplied it, consider whether earlier lessons provided enough inductive data points. (4) Were the DQB orange sticky notes productive (pointing forward to irregular polygons, concave shapes, or coordinate methods) — or superficial? Use productive orange notes to frame the Lesson 5 hook. (5) Equity check: which learners were not cold-called today? Ensure those learners are prioritised in Lesson 5's cold-call rotation. (6) Did the connection to the Kericho land plot feel real and motivating to learners, or did it feel contrived? Note any learner comments about fairness, food security, or land — these are entry points for deeper PCIs integration in future lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners used rulers and triangular card cut-outs to decompose a scaled drawing of the seven-sided Kericho community land plot into five non-overlapping triangles, measuring each triangle's base and perpendicular height.
What did I learn?	The area of a heptagon equals the sum of the areas of five non-overlapping triangles formed by drawing diagonals from one interior vertex: $A = ½b_1h_1 + ½b_2h_2 + ½b_3h_3 + ½b_4h_4 + ½b_5h_5$. In general, any n-gon can be decomposed into (n – 2) triangles.
How does this explain the phenomenon?	Any polygon — no matter how many sides — can be broken into (n – 2) non-overlapping triangles, so its area can always be calculated using the triangle area formula. This means the Kericho community can calculate the exact area of their irregular seven-sided plot and divide it fairly among the three farming families.

LESSON 5 (80 minutes): Octagons — Decomposing the Mombasa Courtyard	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students investigate how the area of a regular octagon can be determined by decomposing it into eight congruent isosceles triangles and by applying a direct formula, then use their result to solve a real-world tiling and costing problem set in a Mombasa courtyard — extending their growing toolkit for finding the area of any polygon, however complex.
Knowledge	Students will know that: (a) a regular octagon has eight equal sides and eight interior angles each measuring 135°; (b) a regular octagon can be decomposed into eight congruent isosceles triangles each with a vertex angle of 45° at the centre; (c) the area of a regular octagon with side length s is $A = 2(1 + \sqrt{2})s^2$; (d) the number of tiles required for a surface equals the total area divided by the area of one tile, and total cost equals number of tiles multiplied by unit price.
Skills	Students will be able to: (a) decompose a regular octagon drawn to scale into eight isosceles triangles and calculate the total area by summing triangle areas; (b) apply the regular octagon area formula $A = 2(1 + \sqrt{2})s^2$ and verify it matches the decomposition result; (c) calculate the number of square tiles needed to tile an octagonal courtyard and estimate the total material cost; (d) update a cumulative area model diagram to include octagons in the decomposition hierarchy.
Attitudes	Students will appreciate that: (a) even shapes that appear complex — like the seven-sided Kericho plot — can always be broken into familiar components whose individual areas can be calculated; (b) mathematical accuracy has direct financial and practical consequences in real Kenyan construction and land-use contexts; (c) working collaboratively and checking answers using two independent methods (decomposition and formula) builds confidence and integrity in mathematical reasoning.
Key Inquiry Question	How do we determine the area of polygons? How is the area of polygons applied in real-life situations?
Purpose in Storyline	In Lesson 5 of the evidence-gathering sequence, students move from heptagons (Lesson 4) to octagons, deepening their understanding that ANY polygon — including the irregular seven-sided Kericho community land plot — can be decomposed into triangles. By mastering the octagon, students add a powerful new case to their cumulative model and consolidate the central insight driving the storyline: decomposition into triangles is a universal area strategy. The Mombasa courtyard tiling problem also introduces financial literacy (cost estimation), showing that precise area calculation has direct economic consequences — just as it does for the farming families dividing the Kericho land plot.
Safety Notes	No significant physical safety concerns for this lesson. Ensure compasses and geometry sets are handled carefully during the construction activity. If printed scale diagrams are used, scissors should be managed safely during any cut-out decomposition tasks. Supervise equitable group participation to maintain a psychologically safe learning environment.

B. LESSON OVERVIEW

Students open with a prediction task about an octagonal courtyard in Mombasa's Old Town and estimate the area before any instruction. They then investigate through two parallel methods — geometric decomposition into eight isosceles triangles and the direct octagon formula — and compare results for consistency. A guided financial literacy extension asks students to calculate tile quantities and total cost in Kenyan Shillings. The lesson closes with a Driving Question Board update and a revision of the cumulative area model diagram (triangles → quadrilaterals → heptagons → octagons), explicitly linking back to the Kericho irregular polygon phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Sinan, Rüstem Pasha Mosque Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student receives a printed card showing a scale diagram of a regular octagonal courtyard in Mombasa's Old Town (side length = 3 m, drawn at 1 cm : 0.5 m). The card states: 'The Mombasa Heritage Trust wants to tile this entire octagonal courtyard with square ceramic tiles (each 0.25 m × 0.25 m). Before we calculate anything, predict: (1) What is your best estimate of the total area of the courtyard in m²? (2) How many tiles do you think will be needed? (3) If each tile costs Ksh 85, estimate the total cost.' Students write individual predictions in their exercise books, then share with a partner (Think-Pair). Three cold-call pairs share predictions aloud; teacher records the range of estimates on the board without judging. Teacher asks: 'How does this problem connect to what the Kericho community is facing with their seven-sided plot?'	Distribute prediction cards as students settle (2 min). Say exactly: 'Look at this octagonal courtyard in Mombasa's Old Town. Without any calculation tools yet — just your mathematical intuition — write down your three predictions. You have 3 minutes of silent individual thinking.' [WAIT — enforce 3 min silence]. Then: 'Now share with your partner for 90 seconds.' After pair-share, cold-call THREE pairs: 'Pair at the front-left window — what area did you estimate and why?' [WAIT 10 sec after question before accepting answer]. Record all estimates on the board in a column. Then ask the whole class: 'Show of hands — who predicted more than 60 m²? Between 40 and 60? Less than 40?' Tally on board. Finally say: 'Keep your predictions visible. By the end of this lesson you will know exactly who was closest — and more importantly, WHY. And think: how is the Mombasa tiling problem connected to the Kericho family land problem?' Cold-call one student for that connection before moving on.	Prediction Anchoring — Students commit to a numerical estimate before instruction, activating prior knowledge of area and creating cognitive dissonance that motivates investigation. Recording the range of class predictions publicly establishes a shared reference point and makes the lesson's resolution (discovering the precise answer) feel meaningful and satisfying.	Observable indicator: every student has written three numerical predictions in their book before pair-share begins. Teacher circulates and scans books during the 3-minute silent phase, noting whether students are drawing on prior area knowledge (e.g., comparing the octagon to a square or rectangle) or writing arbitrary guesses. Students who draw a bounding rectangle around the octagon to estimate are demonstrating productive geometric reasoning and should be noted for cold-calling during Explain phase.
Observe Phase 📹 **VIDEO:** Sinan, Rüstem Pasha Mosque Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for	Students work in groups of four at their desks. Each group receives: (a) a large printed scale diagram of the regular octagon (side = 3 m, scale 1 cm : 0.5 m) with the centre point O marked; (b) a ruler and protractor from the geometry set; (c) a data-collection worksheet. TASK 1 — Decomposition Observation: Students draw straight lines from centre O to each of the eight vertices, dividing	Hand out materials and say: 'Your first job is to be detectives — use your ruler and protractor to investigate the structure of this octagon. Do NOT use any formula yet. Draw the lines from O to each vertex and measure what you find.' [WAIT — allow 8 min of group work without intervention]. Circulate and use targeted prompts only: 'What do you notice about the eight triangles — are	Structured Data Collection with Pattern Recognition — By physically drawing the decomposition lines and measuring angles, students build embodied understanding of WHY the octagon breaks into 8 × 45° isosceles triangles. The pattern table (3 sides → 3 triangles → 120°; 4 sides → 4 triangles → 90°; … 8 sides → 8 triangles → 45°) makes the general principle visible	Observable indicators: (1) All eight lines from O to vertices are correctly drawn on the scale diagram. (2) Groups record a central angle of 45° ± 2° (measurement tolerance). (3) Pattern table entries for octagon are filled in correctly before measuring. Teacher checks the pattern table of at least three groups during circulation. Groups that record an incorrect central

[Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	the octagon into eight triangles. They measure the base (= side length = 3 m) and use the protractor to confirm the central angle of each triangle is 45° (since 360° ÷ 8 = 45°). They measure the height (apothem) of one triangle using their ruler on the scale diagram and record it. TASK 2 — Pattern observation: The worksheet then shows three smaller worked examples of regular polygons decomposed from the centre: equilateral triangle (3 triangles), square (4 triangles), regular hexagon (6 triangles). Students complete the pattern table: 'Number of sides → Number of triangles → Central angle of each triangle'. They predict the entry for the octagon before measuring. TASK 3 — Video/simulation prompt (if device available): Students watch a 90-second animation showing the Mombasa courtyard octagon being 'unfolded' into 8 isosceles triangles and then reassembled, reinforcing the decomposition visually.	they identical?' [WAIT 10 sec]. 'What is the sum of all the central angles? What does that tell you about each one?' After 8 min, pause the class: 'Hands up — what central angle did your group find for each triangle?' Cold-call FOUR groups. Write 45° on the board and derive it explicitly: '360 ÷ 8 = 45°. Now — what type of triangle is this? What do you know about isosceles triangles that will help you find area?' [WAIT 12 sec]. Cold-call TWO students. Then direct groups to Task 2 pattern table. After 4 min, cold-call one group to read their completed table. Close the Observe phase by saying: 'You have now observed the structure. In the next phase you will use what you found to actually calculate the area — two different ways.'	inductively, preparing students for the Explain phase derivation.	angle (e.g., 135°, confusing interior angle with central angle) are flagged for targeted correction during the Explain phase transition.
Explain Phase 📹 **VIDEO:** [Sinan, Rüstem Pasha Mosque] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	PART A — Decomposition Method (Groups): Using their measured apothem from the scale diagram (apothem ≈ 3.62 m for side = 3 m; students derive this from their scale measurement), each group calculates: Area of one isosceles triangle = ½ × base × height = ½ × 3 × 3.62 ≈ 5.43 m². Total octagon area = 8 × 5.43 ≈ 43.5 m². Students record this in their worksheet. PART B — Formula Method (Teacher-led): Teacher writes on the board: $A = 2(1 + \sqrt{2})s^2$ and guides the class through the derivation step-by-step: 'Each isosceles triangle has base s and height (apothem) = $s/2 \times (1 + \sqrt{2})$.	Begin Part A by saying: 'Use your measured apothem from the scale diagram and calculate the area of one triangle, then multiply by 8. You have 5 minutes.' [WAIT — circulate, do not give answers]. After 5 min, cold-call TWO groups for their triangle area and total. Record both on the board. Then say: 'Now I am going to show you a formula that mathematicians derived using exactly the decomposition method you just used.' Write $A = 2(1 + \sqrt{2})s^2$ on the board. Work through the derivation step-by-step, pausing at each line: 'Before I write the next step — what do you think comes next?'	Dual-Method Verification (Decomposition vs. Formula) — By calculating area two independent ways and comparing results, students experience mathematical consistency as a form of proof. This models authentic scientific and engineering practice (triangulating evidence) and builds confidence that the decomposition method is reliable — which is precisely the method that will work on the irregular Kericho polygon where no single formula exists. The financial literacy extension grounds abstract area in concrete Kenyan economic reality.	Observable indicators: (1) Group worksheets show correct triangle area calculation with working (½ × 3 × apothem). (2) Formula substitution is correct: $A = 2(1 + \sqrt{2}) \times 9 = 43.45$ m² (accept 43.2–43.7 m² given measurement variation). (3) Tile count includes the 10% wastage addition. (4) At least one written sentence connecting octagon decomposition to the Kericho problem. Teacher collects one worksheet per group at the end of the phase for rapid review. Students who divide by tile area incorrectly (e.g., forgetting to convert units or not rounding up) are identified for targeted

	Area of 8 triangles = 8 × [½ × s × s/2 × (1 + √2)] = 2(1 + √2)s².' Students substitute s = 3 m: A = 2(1 + 1.414) × 9 = 2 × 2.414 × 9 = 43.45 m². PART C — Verification and Discussion: Students compare their two answers. Teacher asks: 'Why are the two values almost identical? Which method would you trust more and why?' Students write a 2-sentence justification in their books. PART D — Tile and Cost Calculation (Groups): Area of one tile = 0.25 × 0.25 = 0.0625 m². Number of tiles = 43.45 ÷ 0.0625 = 695.2 → round up to 696 tiles. Add 10% wastage (standard construction practice in Kenya): 696 × 1.10 = 766 tiles. Total cost = 766 × Ksh 85 = Ksh 65,110. Groups record and compare their cost estimates with their original predictions. PART E — Kericho Connection: Teacher asks: 'How does what we just did for the octagon help the Kericho community with their seven-sided plot?'	[WAIT 10 sec, cold-call one student each time — target students who were not yet cold-called this lesson]. After Part B substitution, ask: 'Compare your two answers. Are they the same? Why might there be a tiny difference?' [WAIT 12 sec]. Cold-call THREE students. For Part D, say: 'Now you are the contractor hired by the Mombasa Heritage Trust. Calculate exactly how many tiles to order and the total cost in Kenya Shillings. In real construction, you always add 10% for wastage — include that.' [WAIT — allow 6 min group work]. Cold-call ONE group to present their full working. Ask the class: 'Does their answer match yours? If not, where did you differ?' For Part E, ask: 'In one sentence each — how does the octagon decomposition connect back to the Kericho seven-sided plot? Turn and tell your partner.' Then cold-call FOUR students for the connection.		feedback.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Sinan, Rüstem Pasha Mosque Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES	Each student receives two sticky notes (different colours — one for 'What I now understand' and one for 'What I still wonder'). Students write individually for 3 minutes. Sample 'understand' responses expected: 'Any regular polygon can be decomposed into isosceles triangles from the centre,' 'The octagon formula comes directly from the triangle decomposition,' 'Area calculation determines real construction costs.' Sample 'wonder' responses expected: 'Can we decompose an IRREGULAR octagon the same way?', 'What if the sides are all different lengths?', 'How many triangles would the Kericho seven-sided plot split	Distribute sticky notes. Say: 'You have 3 minutes — individual thinking only — no talking. On the YELLOW note write one thing you now understand about area that you did not understand before today. On the GREEN note write the most important question that is still in your mind about the Kericho problem.' [Enforce 3-minute silence — circulate and read notes as students write]. After posting, read THREE green 'wonder' notes aloud (choose ones that point toward irregular polygons and decomposition). Ask: 'Vote with your hand — which of these three questions do you think is MOST important for the Kericho	Driving Question Board Progressive Charting — The public DQB makes student thinking visible and cumulative across all eight lessons. By separating 'what I understand' from 'what I still wonder', students practise metacognitive monitoring. The teacher's act of reading wonder-notes aloud and having the class vote on importance models how mathematicians prioritise open problems — and explicitly connects the octagon lesson back to the overarching Kericho phenomenon.	Observable indicators: (1) Every student has posted two sticky notes. (2) 'Understand' notes reference area, decomposition, or the octagon specifically — not vague statements. (3) 'Wonder' notes point toward the irregular/general polygon problem (Kericho context) rather than unrelated topics. Teacher photographs the DQB at the end of the phase to track conceptual progression across the lesson sequence. Students whose 'understand' notes are still vague ('I understand area') are identified for targeted questioning in the Model Building phase.

for similar readings	into?' Students post their sticky notes on the class Driving Question Board under the columns: 'Shapes we can now find the area of' and 'Questions that will help us solve the Kericho problem.' The teacher reads aloud three 'wonder' notes and asks: 'Which of these questions do you think Lesson 6 will answer?'	community to answer?' Tally votes. Say: 'Look at our DQB — we have now added triangles, quadrilaterals, heptagons and octagons to the shapes we understand. What pattern do you notice in how we solve each one?' [WAIT 12 sec]. Cold-call TWO students. Expected answer: 'We always decompose into triangles.' Affirm: 'Exactly. That principle — decomposition into triangles — is the key that will unlock the Kericho irregular polygon in our final lessons.'		
Model Building Phase 📹 **VIDEO:** Sinan, Rüstem Pasha Mosque Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open their exercise books to their cumulative area model diagram (started in Lesson 1 and updated each lesson). They add a new branch for 'Regular Octagon' showing: (a) the octagon shape labelled with side s and apothem a; (b) an arrow labelled 'decompose from centre' pointing to '8 congruent isosceles triangles, each with central angle 45°'; (c) the formula $A = 8 \times (½ \times s \times a) = 2(1 + \sqrt{2})s^2$; (d) a note: 'Mombasa courtyard: s = 3 m → A ≈ 43.45 m², 766 tiles @ Ksh 85 = Ksh 65,110.' Students then draw a horizontal timeline at the bottom of the model showing the sequence: Triangle (L1) → Rectangle/Parallelogram (L2) → Trapezium (L3) → Heptagon (L4) → Octagon (L5) → [?] (L6, L7) → Irregular Polygon: Kericho Plot (L8). Under the Kericho Plot box, students write: 'Strategy: decompose into triangles — same principle used in ALL previous shapes.' One student per group volunteers to share their updated model diagram with the class using the document camera (or by holding it up). Class gives one	Say: 'Open to your cumulative model diagram. You are going to add today's octagon branch — include the shape, the decomposition arrow, the formula, and the Mombasa numbers. You have 6 minutes.' [WAIT — circulate, do not dictate what to draw. Prompt with: 'Can someone looking at your model diagram understand how the formula was derived?' [WAIT 10 sec]]. After 5 min say: 'Now draw the timeline of all five shapes we have studied. Leave two question-mark boxes for Lessons 6 and 7, and a final box for the Kericho irregular polygon in Lesson 8.' After students complete the timeline, call on ONE volunteer per group (rotate so different students present each lesson) to share their diagram. After each share, ask the class: 'What is one specific thing they included that is mathematically important? What is one addition that would make the model stronger?' [WAIT 10 sec before accepting responses]. Cold-call FOUR students total across the two feedback prompts. Close the lesson by saying: 'Every shape	Cumulative Model Revision — The progressive model diagram is a student-constructed artefact that makes the entire evidence sequence visible in one place. Adding each new polygon as a branch, and drawing the timeline toward the Kericho phenomenon, reinforces that every lesson is purposeful evidence in service of the driving question. Peer feedback using a sentence frame structures collaborative critique without reducing it to praise — modelling how mathematicians and engineers review each other's work.	Observable indicators: (1) Octagon branch is present with shape, decomposition arrow, formula, and numerical Mombasa example. (2) Timeline shows all five shapes in sequence with Kericho polygon at the end. (3) The note under Kericho explicitly references the decomposition-into-triangles strategy. Teacher collects 4–5 model diagrams at random for detailed review before Lesson 6 — checking whether the formula derivation chain (triangle area → × 8 → octagon formula) is shown, not just the formula in isolation. Students whose diagrams show only the formula without the derivation receive written feedback: 'Show WHERE the formula comes from — that is the key to solving the Kericho problem.'

	piece of specific positive feedback and one suggestion for improvement using the sentence frame: 'I notice you included ___. You might also add ___.'	we have studied decomposes into triangles. In Lesson 6 we will see whether that principle holds for a completely irregular polygon — and then in Lesson 8, the Kericho community will finally get their answer.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did students successfully connect the octagon decomposition to the Kericho irregular polygon in Phase 3 Part E and Phase 5? If fewer than 70% of cold-called students made an explicit connection, plan a brief 5-minute bridge activity at the start of Lesson 6. (2) Were the two area methods (decomposition vs. formula) understood as equivalent — or did students treat the formula as a separate 'magic rule'? If so, revisit the derivation in Lesson 6 warm-up. (3) Were any groups unable to read the apothem correctly from the scale diagram? Consider providing a worked measurement example at the start of Lesson 6's Observe phase. (4) Did the financial literacy extension (tile count and cost) generate genuine engagement? Note student reactions — if it was motivating, consider increasing the financial component in Lesson 7. (5) Review the 4–5 collected model diagrams tonight: do they show the derivation chain or only the final formula? Use this to determine the depth of the Lesson 6 warm-up. (6) Review the DQB 'wonder' sticky notes: identify the two or three most common unresolved questions and open Lesson 6 by addressing them explicitly.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that a regular octagon drawn to scale (side = 3 m) can be divided into eight congruent isosceles triangles by drawing lines from the centre O to each vertex. They measured the central angle of each triangle as 45° (confirming 360° ÷ 8 = 45°) and measured the apothem from the scale diagram to use in the triangle area calculation.
What did I learn?	Students learned that the area of a regular octagon can be determined by two equivalent methods: (a) decomposing it into 8 isosceles triangles and summing their areas using $A_{triangle} = \frac{1}{2} \times \text{base} \times \text{height}$; and (b) applying the direct formula $A = 2(1 + \sqrt{2})s^2$. They learned that the formula is derived from the decomposition method, not a separate rule. They also learned how to calculate tile quantities (including 10% wastage allowance) and total material cost in Kenyan Shillings from an area calculation.
How does this explain the phenomenon?	Students explained that the regular octagon's area formula $A = 2(1 + \sqrt{2})s^2$ emerges directly from multiplying the area of one isosceles triangle by eight, and that this decomposition-into-triangles strategy is the same strategy used for every polygon in the lesson sequence (triangles, quadrilaterals, heptagons, octagons). They explained that the same principle — decomposing any polygon into triangles from an interior point — will work for the irregular seven-sided Kericho community land plot, regardless of its irregular side lengths and angles.

LESSON 6 (80 minutes): Irregular Polygons — Triangulation of the Nakuru School Shamba	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to systematically apply the triangulation method to calculate the area of any irregular polygon, and to be introduced to the coordinate (surveyor's) method as an extension technique.
Knowledge	Learners will know that any irregular polygon can be subdivided into non-overlapping triangles using diagonals from a chosen vertex or interior reference point, and that the total area is the sum of the individual triangle areas. Learners will also know the foundational structure of the coordinate/surveyor's shoelace formula.
Skills	Learners will be able to: (a) draw all non-overlapping triangles within a given irregular polygon by selecting a reference vertex or interior point and drawing diagonals; (b) measure base and height (or identify given dimensions) for each triangle; (c) calculate each triangle's area using $A = \frac{1}{2}bh$; (d) sum the triangle areas to find the total polygon area; (e) compare group results, identify sources of measurement discrepancy, and reconcile differences; (f) apply the shoelace formula to verify area using coordinates (extension).
Attitudes	Learners will appreciate that systematic decomposition strategies make seemingly unsolvable real-life measurement problems tractable, and will value precision, collaborative cross-checking, and honest reconciliation of differing answers — mirroring the responsibility required when land decisions affect farming families' food security.
Key Inquiry Question	How do we determine the area of polygons? How is the area of polygons applied in real-life situations?
Purpose in Storyline	In Lessons 1–5 learners built area formulae for triangles, quadrilaterals, regular polygons, and pentagons/hexagons. The Kericho community plot is still unsolved because it has seven irregular sides. Today learners practise on a new, slightly simpler irregular polygon — the five-sided school shamba boundary in Nakuru — using the general triangulation method they have been building toward. Successfully computing this area moves the class one decisive step closer to answering the driving question: they can now tackle the full Kericho seven-sided plot in Lesson 7.
Safety Notes	No chemical or physical hazards. Ensure rulers and geometry sets (compasses, protractors) are handled safely and returned after use. Learners working on the floor with large printed maps should avoid tripping hazards; clear aisle space before the activity begins.

B. LESSON OVERVIEW
Lesson 6 of 8 in the Area of Polygons evidence-gathering sequence. Learners receive a scaled diagram of an irregular five-sided polygon representing the boundary of a school shamba (kitchen garden) near Nakuru. Working in groups, they apply the general triangulation method — choosing a reference vertex, drawing all non-overlapping diagonals to form triangles, measuring each triangle's base and height, and computing the sum of triangle areas to find the total shamba area. Groups then compare answers, identify sources of discrepancy (measurement error, triangle overlap, missed regions), and reconcile to a class consensus. In the final segment, the teacher introduces the coordinate/shoelace formula as an extension tool that surveyors use — connecting explicitly to the Kericho land-division driving question. The lesson advances the driving question by demonstrating that ANY polygon, regardless of irregularity, yields to systematic triangulation.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Regular and Irregular Polygons (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner individually receives a single printed sheet showing: (i) the Kericho seven-sided community plot boundary (from the phenomenon introduction) and (ii) a new five-sided irregular polygon labelled 'Nakuru Shamba Boundary — Scale 1 cm : 4 m'. Without measuring or calculating, learners write in their exercise books: (a) a rough estimate of the shamba's total area in square metres, (b) a prediction of HOW MANY triangles they think can be drawn inside the five-sided shape if they draw lines from one corner, and (c) whether they think their estimate will be an over-estimate or under-estimate once they actually calculate — and why. Learners then share predictions with a shoulder partner for 2 minutes.	Distribute worksheets face-down. Say: 'Do NOT flip your sheet yet.' After 30 seconds of suspense, say: 'Turn over. Look only at the Nakuru shamba — the five-sided shape at the bottom. Before any measuring, write your best estimate of the total area of this shamba in square metres. You have 3 minutes — work alone.' [WAIT 3 minutes silently.] Then: 'Now write: if I draw lines from one corner to every non-adjacent corner inside this shape, how many triangles will I get? Predict and justify.' [WAIT 90 seconds.] Cold-call 4 learners for estimates (select one very low, one very high, two middle estimates). Record all four on the board under 'OUR PREDICTIONS'. Ask: 'Why do our estimates differ so much? What does that tell us about measuring irregular shapes by eye?' [WAIT 10 seconds.] Accept 2–3 responses. Then say: 'Today we will find the exact answer using a method that works for ANY polygon — and by the end of this lesson, you will be ready to solve the Kericho plot.'	Prediction with justification (Predict-then-Justify): By committing to an estimate and a reasoning before instruction, learners activate prior area knowledge, surface misconceptions about irregular shapes, and create cognitive investment in the outcome. Recording all estimates publicly ensures the Observe phase has a concrete target to confirm or refute.	Circulate during the 3-minute prediction window. Observable indicators: (1) Learner can write a numerical estimate with square-metre units — if a learner writes only a bare number without units, probe: 'What units should area be in?' (2) At least 3 of 4 cold-called learners predict 3 triangles from one vertex of a pentagon — this reveals whether learners recall or can reason the (n−2) triangle rule. (3) Listen for the phrase 'I think it will be an over-estimate because I rounded up' — this shows proportional reasoning. Flag learners who leave parts (b) or (c) blank for targeted support in Phase 2.
Observe Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Regular and Irregular Polygons (Flexbook)]	Groups of 4 receive: the same scaled shamba worksheet (pentagon, vertices labelled A–B–C–D–E, grid lines at 1 cm intervals, scale 1 cm : 4 m), a geometrical set (ruler, protractor, set square), and a group recording table with columns: Triangle Name | Base (cm) | Height (cm) | Area (cm^2) | Area (m^2). Step 1 — Triangulation: Each group chooses vertex A as the reference point and draws diagonals AC and AD, partitioning the pentagon into triangles ABC, ACD, and ADE. Learners sketch the triangulation	Before groups begin, hold up the pentagon worksheet and ask: 'How many triangles do we expect from vertex A in a five-sided polygon? Use the rule we discussed.' [WAIT 10 seconds.] Cold-call 2 learners — confirm answer is 3 (= n − 2 = 5 − 2). Say: 'Your job is to decompose, measure, compute, and total. Every person in the group MUST measure at least one triangle. Record everything — including any measurement you are unsure about.' While groups work (~20 minutes), circulate using these specific probing questions:	Structured group decomposition with public evidence posting: Breaking the task into explicit steps (triangulate → measure → calculate → sum → post) mirrors the scientific practice of Planning and Carrying Out Investigations (NGSS SEP 3). Public posting of all group answers turns individual results into class-level data, making discrepancies visible and motivating the reconciliation discussion in Phase 3.	Observable indicators during circulation: (1) All groups correctly draw exactly 3 non-overlapping, non-boundary-crossing diagonals from vertex A — if a group draws only 2 diagonals, their polygon is not fully triangulated; intervene immediately. (2) Each learner can state the area formula A = ½bh without prompting — if a learner writes A = b × h, redirect: 'What shape is this? How does the triangle formula differ from the rectangle formula?' (3) Conversion from cm^2 to m^2 is correct: since scale is 1 cm : 4 m, 1 cm^2 = 16 m^2

Source: CK-12 🔍 Search ARES for similar readings	on their worksheet, clearly labelling each triangle. Step 2 — Measure: Each group member is assigned one triangle and measures its base and perpendicular height using ruler and set square. All measurements are recorded in the table. Step 3 — Calculate: Each learner computes area of their assigned triangle (½ × base × height), converts from cm² to m² using the scale factor (1 cm² = 16 m²), and enters the result. Step 4 — Sum: The group sums all three triangle areas to find total shamba area in m². Step 5 — Cross-check: Each group writes their total on a sticky note and posts it on the 'Evidence Wall' at the front of the room.	(1) To any group drawing diagonals incorrectly: 'Does that diagonal stay inside the polygon? Check if it crosses a boundary line.' (2) To any group measuring slant height: 'Is that the PERPENDICULAR height? Use your set square to confirm it meets the base at 90°.' (3) To a fast-finishing group: 'Try choosing vertex C as your reference point instead of A. Do you get the same three triangles? Will the total area change?' After all sticky notes are posted, pause the class: 'Look at the Evidence Wall. Are all totals the same? Which groups are close? Which are far apart? We will investigate why in Phase 3.'		— a common error is multiplying by 4 instead of 16; identify these groups for targeted Phase 3 correction. (4) Sticky-note totals: teacher notes the range of posted answers to calibrate the depth of reconciliation needed.
Explain Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Regular and Irregular Polygons (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class gallery walk (3 minutes): All learners walk past the Evidence Wall and record the lowest and highest group total in their books. Return to seats. Class discusses: Why do different groups get different answers for the SAME shape? Learners identify three possible reasons: measurement error (ruler misread), perpendicular height error (slant used instead of true height), and conversion error (scale factor applied incorrectly). Teacher facilitates a step-by-step worked reconciliation on the board using one group's data, correcting each error type. Learners recalculate their own group total if needed and agree on a class consensus value. Extension segment (~10 minutes, flagged as 'Challenge for Fast Finishers and Future Surveyors'): Teacher introduces the coordinate/shoelace formula. The five pentagon vertices are given coordinates on a	After gallery walk, ask: 'Hands up if your group's answer is within 5 m² of the class highest value? Within 20 m²? More than 20 m² away?' Count hands publicly — this creates a self-assessment moment. Then say: 'Let us diagnose the errors systematically. I will use Group 3's numbers — Group 3, bring your worksheet to the visualiser.' Work through on board: (1) 'Group 3 measured this height as 3.8 cm — let me show you why it should be 3.2 cm using the set square.' (2) 'Group 3 converted area by multiplying by 4 — can someone explain why we must multiply by 16?' [WAIT 10 seconds.] Cold-call 1 learner. (3) Confirm corrected total. Say: 'Everyone, re-check your group's conversion step right now — multiply each cm² area by 16, not 4.' [Allow 2 minutes.] Then connect to the phenomenon: 'If the Kericho elder used the wrong scale factor, the land division would be unfair.	Error-diagnosis reconciliation followed by expert-method extension (Constructing Explanations — NGSS SEP 6): Reconciling discrepancies through systematic error-type diagnosis deepens understanding of WHY precision matters, not just HOW to calculate. Introducing the shoelace formula as a 'professional surveyor's tool' positions mathematics as a living practice used by Kenyan professionals, bridging the classroom to real land administration contexts at the National Land Commission.	Observable indicators: (1) After reconciliation, every group's written total should fall within ±8 m² of the class consensus — if a group is still outside this range, identify their specific step error. (2) When asked 'why do we multiply by 16 not 4?', at least 4 learners should be able to explain: 'because area scales by the square of the linear scale factor ($4^2 = 16$)' — this is a key conceptual checkpoint connecting to earlier Sub-Strand 2.1 (Similarity and Enlargement). (3) Extension learners who apply the shoelace formula: check that they correctly alternate sign of products ($x_iy_{i+1} - x_{i+1}y_i$) and take the absolute value — a correct final answer within ±1 cm² of the triangulation answer confirms mastery.

	grid: A(0,0), B(3,0), C(5,2), D(4,5), E(1,4) (in cm on the scale diagram). Teacher writes the shoelace formula on the board: Area = $\frac{1}{2}\lvert\Sigma(x_i y_{i+1} - x_{i+1} y_i)\rvert$. Teacher demonstrates ONE full worked example with these coordinates, showing the tabular method. Learners who finish early verify their triangulation answer using the shoelace result.	Precision matters for people's livelihoods.' For the extension, say: 'Surveyors in Kenya, including those at the National Land Commission, use a formula called the Shoelace Formula — it skips all the triangle drawing. Watch me apply it to our exact pentagon.' Demonstrate the tabular shoelace calculation step by step. Ask: 'Does the shoelace answer match our triangulation answer?' [WAIT 10 seconds.] Cold-call 1 learner to confirm. Say: 'This is one more tool we can use to verify our Kericho plot calculation next lesson.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Regular and Irregular Polygons (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group reviews the class Driving Question Board. They write responses to two prompts on colour-coded sticky notes: GREEN note — 'What new evidence from today helps us solve the Kericho land-division problem?' (e.g., 'We can break the 7-sided plot into 5 triangles using diagonals from one vertex'); ORANGE note — 'What do we still need to figure out before we can fairly divide the Kericho plot?' (e.g., 'We need the actual measurements of each side and diagonal of the Kericho plot' or 'We need to confirm the scale factor for the Kericho boundary drawing'). Groups post their sticky notes under the DQB columns 'Evidence We Now Have' and 'Still Need to Investigate'. The class reads the new notes aloud (rotating reader, one note per group) and the teacher annotates the DQB to show the cumulative progress across Lessons 1–6.	Say: 'We have been building toward one goal since Lesson 1 — solving the Kericho community land problem. Take 4 minutes: your group writes one GREEN note and one ORANGE note for the DQB.' [WAIT 4 minutes.] Circulate and prompt any group writing a vague GREEN note: 'Can you be more specific — how exactly does today's method apply to the seven-sided Kericho plot?' After posting, rotate reader: 'Amara, read Group 2's green note aloud. Does everyone agree this is valid evidence?' After all notes are read, draw a large arrow on the DQB connecting today's lesson marker to the Lesson 7 slot and say: 'Next lesson we go back to Kericho — seven sides, three families, equal shares. You now have the tool. What remains is applying it carefully.' Ask one final cold-call question to the class: 'If we can break any polygon into triangles, does our method work for the Kericho plot even if its sides are all different lengths?' [WAIT 10 seconds.] Accept 2 responses —	Cumulative DQB annotation (Obtaining, Evaluating, and Communicating Information — NGSS SEP 8): The DQB makes the class's growing understanding visible as a shared artefact. Separating 'Evidence We Now Have' from 'Still Need to Investigate' mirrors scientific practice, helping learners distinguish established claims from open questions — a metacognitive habit that supports independent problem-solving on the Kericho plot.	Observable indicators: (1) GREEN notes contain specific methodological claims ('triangulate from vertex A into $n-2$ triangles', 'convert area using scale²') rather than vague statements ('we learned about triangles') — vague notes indicate surface-level processing; prompt those groups to revise. (2) ORANGE notes reveal genuine remaining gaps: if a group writes 'we need to know what a polygon is', this indicates they have not consolidated prior lessons and need targeted review before Lesson 7. (3) Teacher tallies how many groups independently mention the scale factor squared — this informs whether a brief recap is needed at the start of Lesson 7.

		confirm: 'Yes — the triangulation method does not require any equal sides or regular angles. That is its power.'		
Model Building Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Regular and Irregular Polygons (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually update their 'Area of Any Polygon' model — a cumulative diagram they have been building since Lesson 1 in a dedicated section of their exercise books. Today they add: (a) a labelled sketch of the Nakuru pentagon showing the three triangulation diagonals from vertex A, with each triangle shaded a different colour and labelled with its area; (b) a generalisation statement written in their own words: 'To find the area of any n-sided polygon, draw (n−2) non-overlapping triangles by connecting one vertex to all non-adjacent vertices. Sum all triangle areas.'; (c) an arrow from their pentagon model pointing to a small sketch of the Kericho seven-sided plot with the annotation: 'Lesson 7: n = 7, so I will draw ____ triangles.' Learners who completed the extension also add a summary box: 'Shoelace/Surveyor's Method: works directly from vertex coordinates — useful for Kenya National Land Commission surveys.' At the very bottom of their model page, learners write one sentence connecting today's work to the phenomenon: 'Today's method can be applied to the Kericho plot because ___.'	Say: 'Open your model-building page. You have 8 minutes to update your cumulative model with today's three additions — the pentagon sketch with shaded triangles, the generalisation statement, and the Lesson 7 prediction.' [WAIT 8 minutes — circulate silently, reading over shoulders.] After time is up, ask: 'How many triangles will you draw for the Kericho seven-sided plot in Lesson 7?' [WAIT 5 seconds.] Cold-call 3 learners — confirm answer is 5. Then ask: 'If each triangle has an average area of 120 m², give me a rough estimate of the total Kericho plot area.' [WAIT 10 seconds.] Cold-call 2 learners — accept any answer near 600 m² with correct reasoning (5 × 120). Say: 'Hold onto that estimate — we will check it against the real calculation next lesson. For homework, re-draw the Kericho plot from Lesson 1's worksheet, choose your reference vertex, draw all five diagonals, and label each of the five triangles. Do not calculate yet — just set up the triangulation. Bring it to Lesson 7.'	Cumulative model revision with explicit forward-linking (Developing and Using Models — NGSS SEP 2): Each lesson's model update deepens the explanatory diagram rather than replacing it. Asking learners to annotate the Kericho plot with a prediction ('I will draw ____ triangles') creates a bridge structure so that Lesson 7 begins with an actionable plan already in place, reducing cognitive load at the point of application.	Observable indicators: (1) The generalisation statement uses the term 'n−2 triangles' or an equivalent correct expression — if a learner writes 'n triangles', flag for correction. (2) The Lesson 7 prediction correctly states 5 triangles for the seven-sided Kericho plot — error here means the (n−2) rule has not been retained; re-teach at the start of Lesson 7. (3) The phenomenon-connection sentence contains a logical 'because' clause linking the method to the polygon's properties (e.g., 'because the Kericho plot can also be divided into 5 non-overlapping triangles from one vertex, and we just showed this works for any polygon') — a sentence without a 'because' clause suggests declarative but not explanatory understanding.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) TRIANGULATION ACCURACY — Did most groups correctly identify exactly 3 non-overlapping triangles in the pentagon, or did several groups draw diagonals that crossed outside the polygon boundary? If more than 2 groups struggled with interior diagonal placement, plan a brief 5-minute visual demonstration at the start of Lesson 7 using a cut-out polygon on the board. (2) SCALE FACTOR CONVERSION — Did learners correctly apply the squared

scale factor (×16) or did many multiply by 4? This error is a critical gap that will corrupt all Kericho plot calculations in Lesson 7; if more than one-third of groups made this error, open Lesson 7 with a 3-question warm-up exclusively on area scale factors, connecting to Sub-Strand 2.1 (Similarity and Enlargement). (3) RECONCILIATION QUALITY — Did the whole-class reconciliation genuinely converge to a consensus, or did some groups remain unconvinced? If the Evidence Wall sticky notes showed a very wide spread (>30 m² range), consider assigning a short individual recalculation task as homework using a simpler 4-sided polygon to consolidate measurement technique. (4) EXTENSION ENGAGEMENT — How many learners engaged meaningfully with the shoelace formula? If fewer than 20% attempted it, consider making it a paired challenge in Lesson 7's warm-up rather than a Phase 3 extension. (5) DQB QUALITY — Were the ORANGE 'still need to investigate' notes specific enough to drive Lesson 7 planning? If notes were vague, open Lesson 7 by re-reading them as a class and sharpening them together before beginning the Kericho calculation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners worked with a scaled diagram of a five-sided irregular polygon representing the boundary of a school shamba near Nakuru. They drew three non-overlapping triangles inside the pentagon by connecting vertex A to all non-adjacent vertices, measured each triangle's base and perpendicular height using rulers and set squares, computed each triangle's area using $A = \frac{1}{2}bh$, converted from cm² to m² using the squared scale factor (1 cm² = 16 m²), and summed the three triangle areas to find the total shamba area. Groups posted their totals on the Evidence Wall and observed that different groups obtained different answers for the same shape.
What did I learn?	Any irregular polygon with n sides can be divided into exactly (n−2) non-overlapping triangles by drawing diagonals from one reference vertex to all non-adjacent vertices. The total area of the polygon equals the sum of the areas of these triangles ($A = \frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 + \ldots + \frac{1}{2}b_{n-2}h_{n-2}$). When working from a scale diagram, areas must be converted using the square of the linear scale factor (if 1 cm represents k metres, then 1 cm² represents k² square metres). Discrepancies between group results arise from measurement error or incorrect perpendicular height identification, not from the method itself. The coordinate (shoelace) formula provides an alternative verification method used by professional surveyors.
How does this explain the phenomenon?	The triangulation method works for ANY polygon — regular or irregular, regardless of the number of sides — because every polygon can be decomposed into non-overlapping triangles, and we already know how to find the area of a triangle. This means the method discovered today can be applied directly to the seven-sided Kericho community plot: by drawing five diagonals from one vertex, we create five triangles whose areas sum to the total plot area, giving the village elder the precise measurement needed to divide the land fairly among the three farming families.

LESSON 7 (80 minutes): Composite Polygons and Real-Life Application: Solving the Kericho Land-Division Problem	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to calculate the area of composite polygons by combining and subtracting standard polygon areas, and to apply this skill to solve the anchoring phenomenon — fairly dividing the Kericho community land plot among three farming families.
Knowledge	Learners know and understand: (i) that a composite polygon is a shape made up of two or more standard polygons combined or one subtracted from another; (ii) how to calculate areas of triangles, rectangles, parallelograms, trapezoids, and irregular polygons using previously established formulae; (iii) that the total area of a composite polygon equals the sum of its component areas, or a whole area minus a removed section; (iv) that the Kericho seven-sided plot can be decomposed into triangles and quadrilaterals whose areas can be summed to give a total, which can then be divided equally into three parts.
Skills	Learners are able to: (i) decompose a composite polygon into recognisable standard shapes and label all necessary dimensions; (ii) calculate the area of each component shape accurately and sum or subtract to find total area; (iii) divide a total polygon area equally into three sections and justify the division with labelled diagrams; (iv) communicate a complete, justified solution to the Kericho land-division problem in writing and orally to peers.
Attitudes	Learners appreciate the social importance of accurate area calculation in land use, food security, and community fairness; they value mathematical precision as a tool for equity and responsible citizenship; they demonstrate perseverance in multi-step problems and respect for peers' solution strategies.
Key Inquiry Question	1. How do we determine the area of polygons? 2. How is the area of polygons applied in real-life situations?
Purpose in Storyline	Lesson 7 is the culminating evidence-gathering lesson. Learners have spent Lessons 1–6 building a toolkit: triangle areas, quadrilateral areas (rectangles, parallelograms, trapezoids), regular polygon areas, and decomposition of irregular polygons. In this lesson they first consolidate the combination/subtraction technique on two warm-up composite shapes (an L-shaped school building footprint and a pentagonal community garden with a rectangular section removed), then turn that full toolkit on the anchoring phenomenon itself — the Kericho seven-sided plot. Groups calculate the total plot area, propose a fair three-family division, and produce a labelled diagram with a written justification. This sets up Lesson 8, where groups will present and critique each other's solutions and agree on a community recommendation.
Safety Notes	No physical hazards. Ensure rulers and geometrical sets are handled carefully. If using printed scaled diagrams, scissors should be used under teacher supervision only during the optional physical decomposition activity.

B. LESSON OVERVIEW

Lesson 7 is an 80-minute composite-polygon problem-solving and application lesson. It opens with a 10-minute Predict Phase in which learners revisit the Kericho plot boundary (drawn to scale on a worksheet) and predict — before calculating — how many standard shapes they can see hidden inside the seven-sided polygon and roughly what the total area might be. In the 20-minute Observe Phase, groups tackle two structured composite-polygon problems (the L-shaped Kiambu school building footprint and the Nakuru pentagonal community garden), recording their area-calculation steps systematically. The 20-minute Explain Phase uses a Gallery Walk sensemaking strategy: groups post their workings, rotate, leave sticky-note critiques, and consolidate a whole-class agreed method for combining and subtracting polygon areas. The 20-minute Driving Question Board phase returns learners to the Kericho phenomenon: groups apply their full toolkit to calculate the total land area and propose a fair three-family division, producing a labelled diagram and written justification. The final 10-minute Model Building Phase has each group revise their running cumulative model (started in Lesson 1) to incorporate the composite-polygon principle and add the Kericho solution, ready for the Lesson 8 presentation.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prime and Composite Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a scaled worksheet showing the Kericho seven-sided plot boundary (to scale, with all side lengths and one diagonal labelled). Working individually for 3 minutes, each learner studies the boundary and sketches on the worksheet any internal lines they would draw to break the plot into shapes they already know — triangles, rectangles, trapezoids. They write two predictions on a sticky note: (1) 'I think the plot is made up of ___ standard shapes,' and (2) 'My rough estimate of the total area is ___ m².' Learners also receive a second worksheet showing an L-shaped building footprint (the Kiambu Girls' Secondary School new science block) and a pentagonal garden outline (Nakuru Community Organic Garden) with a rectangular vegetable patch removed from its interior. They predict which composite problem will be easier to solve and why, writing one sentence of reasoning. Predictions are posted on a class Predict Wall beside the Driving Question Board.	Distribute the scaled Kericho plot worksheet and the two composite warm-up worksheets. Say: 'Before we calculate a single number, I want your best mathematical instinct. Look at this boundary — the same one the Kericho village elder drew. Sketch any internal lines that would help you find the area. Then write your two predictions on your sticky note. You have 3 minutes of silent, individual thinking.' Set a visible timer. After 3 minutes, cold-call 4 students (choose one from each corner of the room) to share their sketched decompositions on the board — do NOT affirm or correct yet. Ask: 'Does anyone see a decomposition that is different from all four on the board?' (Wait 10 seconds.) Probe: 'Baraka, you drew a diagonal from vertex A to vertex E — what was your reasoning?' (Wait 12 seconds for response.) Say: 'Hold your predictions — we will return to this plot in Phase 4 with the tools we sharpen in Phases 2 and 3.'	Prediction with Decomposition Sketching — Learners activate prior knowledge of standard polygon areas (built in Lessons 1–6) by physically marking how they would break the irregular boundary into known shapes. Sketching forces commitment to a decomposition strategy before calculation, creating productive cognitive dissonance when groups later discover multiple valid decompositions yield the same total area — a key insight that deepens understanding of additivity of area.	Observable indicator: Each learner has produced a labelled decomposition sketch and two written predictions. Teacher scans sticky notes during cold-call for: (i) whether learners are drawing triangles and quadrilaterals (correct approach) or trying to fit a single formula (misconception to address); (ii) whether rough estimates show number sense (e.g., a plot labelled ~30 m × 20 m should yield estimates in the hundreds of m², not thousands). Flag any learner who has left the worksheet blank — pair them with a peer during Phase 2.
Observe Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners work in their established groups of four. Each group receives: (a) the L-shaped Kiambu school building footprint worksheet (an L-shape with all outer dimensions labelled: overall length 18 m, overall width 12 m, the rectangular notch cut out is 6 m × 7 m); (b) the Nakuru pentagonal community garden worksheet (a	Circulate continuously. At the 5-minute mark, stop at two groups: (1) a group that subtracted the notch from a full rectangle — ask 'Can you show me which rectangle you started with? What were its dimensions?' (Wait 10 seconds.); (2) a group that split the L into two rectangles — ask 'How did you decide where to make the cut?	Structured Evidence Table with Parallel Decomposition — The three-column recording table forces learners to name each sub-shape, state its dimensions, and calculate its area before summing, mirroring the scientific practice of systematic data recording (NGSS SEP 3: Planning and Carrying Out Investigations, adapted for	Observable indicators: (i) Evidence tables are fully populated — no blank 'Dimensions' cells; (ii) Both decompositions of the L-shape give 156 m² (18×12 − 6×7 = 216 − 42 = 174 m² — NOTE: teacher checks own answer key; the correct value depends on the exact worksheet dimensions provided, so teacher must verify

📖 **HTML:** Prime and Composite Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	rectangle 15 m × 10 m with a triangular section of base 5 m and height 4 m added to one short side, and a rectangular vegetable patch 3 m × 4 m removed from the interior). Groups are given a structured evidence-recording table with three columns: 'Shape name,' 'Dimensions used,' 'Area calculated.' They must complete the table for each composite figure before writing the total area. Groups are explicitly instructed: 'For the L-shape, use at least TWO different decomposition strategies and confirm both give the same total.' After 15 minutes of group work, each group writes their two final answers (L-shape total area and garden total area) on a mini-whiteboard or large paper and holds it up simultaneously on the teacher's signal.	Could you cut it a different way and still get the same answer?' (Wait 12 seconds.) At the 10-minute mark, call a 30-second whole-class pause: 'Raise your hand if your two decomposition strategies for the L-shape gave the same total. (Scan room.) If they did not match, check your arithmetic — the areas MUST match; that is a mathematical law, not a coincidence.' For the pentagonal garden, probe: 'Wanjiru, your group subtracted the vegetable patch — what does that mean physically for the Nakuru farmers? What part of the garden is left for planting sukuma wiki?' (Wait 10 seconds.) Signal groups to hold up mini-whiteboards at 15 minutes. Say: 'I can see most groups agree on 156 m² for the L-shape and 147 m² for the garden. Two groups have different values — we will find out exactly where the difference lies during the Gallery Walk.'	mathematics). Requiring two decomposition strategies for the L-shape builds understanding that area is an intrinsic property of a shape, independent of the strategy used to calculate it — a foundational mathematical idea that directly supports the Kericho problem, where any valid decomposition of the seven-sided plot must yield the same total area.	before class); (iii) Garden total is computed as (15×10) + (½×5×4) − (3×4) = 150 + 10 − 12 = 148 m²; (iv) Groups that have wrong answers have arithmetic errors in one sub-shape, not a conceptual error — teacher notes these groups for targeted support during Phase 3.
Explain Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prime and Composite Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups post their completed evidence tables and worked solutions for both composite problems on the classroom walls (Gallery Walk setup). Each group's paper must include: the decomposition diagram (labelled), the evidence table, and the final area. Groups then rotate in a set direction, spending 3 minutes at each of three other groups' posters. At each poster, they place a green sticky note if they agree with the solution and can confirm the arithmetic, or a yellow sticky note if they found an error or used a different decomposition that gave a different answer, writing the specific step where they diverged. After rotation (12 minutes), groups	Before rotation begins, say: 'Your job at each poster is not to admire — it is to audit. Check every number. If you find a step you cannot follow, that is a yellow note. If you confirm everything, green note.' During rotation, position yourself near the posters with flagged discrepancies from Phase 2. After rotation, during the 3-minute defence window, cold-call one group that received a yellow note: 'Kamau, your group got a yellow note on the garden problem. Read it out. Do you agree with the critique?' (Wait 12 seconds.) If error confirmed: 'Where exactly did the arithmetic diverge? Can you fix it now in front of us?' For synthesis, write on the	Gallery Walk with Sticky-Note Audit — This peer-critique strategy (aligned with NGSS SEP 7: Engaging in Argument from Evidence) requires learners to evaluate mathematical reasoning, not just answers. Placing sticky notes forces explicitness: a learner cannot simply say 'it looks wrong' — they must identify the specific step. The 3-minute return-and-defend window builds mathematical argumentation skills directly relevant to Phase 4, where groups must justify their Kericho solution to the community (represented by classmates in Lesson 8). The teacher's synthesis bridges the warm-up context (school building, community	Observable indicators: (i) Every group's poster has received at least one sticky note (green or yellow); (ii) Groups that received yellow notes are able to identify the specific arithmetic or conceptual error within the 3-minute window — if a group cannot find their error, teacher intervenes with a direct question: 'What is the area of just this rectangle here? Read the dimensions off the diagram.'; (iii) During synthesis cold-calls, at least 2 of 3 students can articulate the addition/subtraction principle in their own words without reading from the board — this is the minimum threshold to proceed to Phase 4.

	return to their own poster, read the feedback, and have 3 minutes to either defend their answer or correct an error. The whole class then converges for a 5-minute teacher-led synthesis: the teacher writes the generalised composite-polygon rule on the board — 'Total Area = Sum of component areas (addition) OR Whole area minus removed area (subtraction)' — and connects it explicitly to the Kericho plot: 'The Kericho plot is a composite polygon. The same rule you just applied to the school building and the community garden applies directly to the village elder's boundary.'	board: 'Area(composite) = Area(shape 1) + Area(shape 2) + … ± Area(removed section).' Ask the whole class: 'How does this single equation connect to everything we have done in Lessons 1 through 6?' (Wait 15 seconds.) Cold-call 3 students. Then say: 'In Phase 4, you will apply exactly this equation to the Kericho plot. Your Phase 1 decomposition sketch is your starting point — but you may now improve it.'	garden) to the anchoring phenomenon, maintaining storyline coherence.	
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Prime and Composite Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups return to the scaled Kericho plot worksheet from Phase 1. Each group now has 15 minutes to produce a complete, justified solution to the anchoring phenomenon: (1) Decompose the seven-sided plot into standard shapes using internal construction lines, labelling every sub-shape with its name, dimensions, and calculated area. (2) Sum the component areas to find the total plot area in m². (3) Divide the total area by 3 to find the equal share for each of the three farming families (the Wanjiku family for tea, the Otieno family for maize, and the Kosgei family for mixed crops). (4) On the same diagram, shade and label the three sections with each family's name and area. (5) Write two sentences justifying why the division is fair (equal area) and whether the shape of each section is practical for farming. At 15 minutes, each group nominates one spokesperson. Spokespersons stand and give a 30-second summary of their	Re-distribute Phase 1 worksheets (or fresh copies of the scaled Kericho plot). Say: 'You have 15 minutes. Everything you have built in Lessons 1 through 7 leads to this moment. The Wanjiku, Otieno, and Kosgei families are waiting for your answer. Show every step — a correct answer with no working cannot be trusted in a land dispute.' Circulate and check: at 5 minutes, verify that every group has drawn internal construction lines and is not trying to apply a single formula. If a group is stuck, ask: 'Achieng, which diagonal did you sketch in Phase 1? Start there — what two shapes does it create?' (Wait 10 seconds.) At 10 minutes, check that at least one group member is working on the equal-division calculation. Probe: 'If the total is 840 m², how much does each family get? Can you mark that boundary on your diagram?' After spokespersons summarise, facilitate the thumb vote. Do NOT announce a 'correct' total area — say: 'We have two	Anchored Application with Public Commitment — Returning to the exact phenomenon introduced in Lesson 1 with a full solution demand (rather than a sub-problem) creates meaningful closure and motivates precision. The 30-second public summary and thumb vote is a low-stakes public commitment strategy: by committing to a total area in front of peers, groups are motivated to check their arithmetic and reasoning. The DQB update makes the growth of collective understanding visible and tangible — learners can see the journey from 'how do we find triangle areas?' (Lesson 1) to 'we can now solve any polygon area problem' (Lesson 7), which reinforces self-efficacy.	Observable indicators: (i) Every group has a labelled decomposition diagram with at least 3 named sub-shapes, individual areas calculated, and a summed total; (ii) Every group has computed Total ÷ 3 and shaded three sections on the diagram; (iii) Spokesperson summaries include a stated total area and a stated per-family area — vague answers ('we got a big number') are redirected: 'Give us the number in m², Njoroge'; (iv) The class thumb vote reveals whether a consensus total area is emerging — if more than 2 groups diverge significantly (>5% difference), teacher notes this as the opening problem for Lesson 8 peer-review.

	group's total area and proposed division. The class votes with thumbs: up (agree with total area), sideways (unsure), down (disagree with total). Groups with thumbs-down neighbours must note the discrepancy for Lesson 8. The teacher then updates the class Driving Question Board: moves the card 'Can any polygon be broken into shapes we know?' to the 'We Now Know' column, and adds a new card: 'Does the shape of each family's section affect its farming usefulness? (Lesson 8).'	groups with 840 m² and one with 856 m². In Lesson 8 we will find out where the 16 m² difference comes from, and the community will need to trust the most carefully justified calculation.' Update the DQB visibly and say: 'What question does this raise about the shapes of the three sections?' (Wait 12 seconds.) Cold-call 2 students.		
Model Building Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prime and Composite Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner opens their individual running cumulative model — a diagrammatic record updated at the end of every lesson since Lesson 1 that shows how their understanding of polygon area has grown. In Lessons 1–3 it showed triangles and basic quadrilaterals; in Lessons 4–5 it grew to include trapezoids and parallelograms; in Lesson 6 it incorporated irregular polygon decomposition. Now, in 8 minutes, each learner adds a new section to their model titled 'Composite Polygons — Lesson 7.' In this section they must: (i) draw and label one example of an additive composite polygon (like the L-shaped building) and one example of a subtractive composite polygon (like the pentagonal garden); (ii) write the general rule: Area(composite) = Σ Area(components) ± Area(removed section); (iii) draw a small thumbnail of the Kericho plot with their chosen decomposition and write their group's total area. In the final 2 minutes, learners write one 'remaining question' on a sticky note and add it to the DQB. Examples the teacher anticipates:	Say: 'Your model is the story of your mathematical thinking across this entire unit. Lesson 7 is the moment the story comes together — composite polygons are not a new topic; they are everything we have learned, combined. In 8 minutes, add the Lesson 7 section. Be precise with labels and make the general rule prominent — you will need it in Lesson 8 when we defend our Kericho solutions.' Circulate and look for models that show only the additive case — prompt: 'Amina, your model shows combining shapes. Can you also show me an example where you remove a shape? The Nakuru garden from today is perfect.' At the 8-minute mark, say: 'One minute — write your remaining question on a sticky note.' After sticky notes are posted, read 2–3 aloud without attribution: 'Someone asks: what if we cannot measure the internal heights? Excellent — that is exactly what trigonometry, which you studied in Sub-Strand 2.4, helps us with. We will explore connections in Lesson 8.' Close by saying: 'The Kericho families are depending on	Cumulative Model Revision — The running individual model is the signature sensemaking artefact of this lesson sequence. Requiring learners to integrate today's composite-polygon learning into a model that already contains Lessons 1–6 content forces them to see the coherent structure of the sub-strand: each lesson has added a tool to the toolkit, and Lesson 7 shows how all tools combine to solve the anchoring phenomenon. Writing the remaining question externalises metacognition and feeds directly into Lesson 8's agenda, maintaining student ownership of the inquiry.	Observable indicators: (i) Model section for Lesson 7 contains both an additive and a subtractive example with labelled dimensions and correct areas; (ii) General rule is written symbolically (not just in words); (iii) Kericho thumbnail is present with a total area figure; (iv) Remaining question is specific and mathematically meaningful — vague questions ('I don't understand') are addressed one-on-one: teacher asks 'What specifically is confusing — the decomposition step, the arithmetic, or the division into thirds?' Exit check: teacher collects 5 randomly chosen models at the door and reviews them before Lesson 8 to identify learners who need targeted support during the presentation phase.

	'What if we can't measure the height of a triangle inside the plot?', 'Is there a single formula for any polygon?', 'What if the land is on a slope?'	mathematicians like you. Come to Lesson 8 ready to present, defend, and agree on a final community recommendation.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. DECOMPOSITION DIVERSITY — Did groups use different valid decompositions for the Kericho plot? If all groups used the same decomposition, was it because they shared ideas too early, or because the diagram made one decomposition too obvious? Consider whether the Phase 1 worksheet should have fewer visual cues (e.g., remove the pre-labelled diagonal) in future iterations.

2. ARITHMETIC ACCURACY — How many groups arrived at the same total area for the Kericho plot? If fewer than 70% of groups agreed, was the discrepancy due to: (a) arithmetic errors in sub-shape areas (most likely), (b) different but both valid decompositions producing rounding differences (acceptable), or (c) a genuine conceptual error (e.g., double-counting a shared triangle)? Note the specific error pattern for Lesson 8's opening discussion.

3. EQUAL DIVISION — Did any group notice that dividing total area equally into three does not automatically produce three sections with convenient boundaries on the original plot? This is a richer mathematical insight — if it emerged, honour it explicitly in Lesson 8. If it did not emerge, plan a probing question for Lesson 8: 'Is a section of land that is mathematically equal in area always practically equal for farming?'

4. KENYAN CONTEXT ENGAGEMENT — Did the Wanjiku/Otieno/Kosgei family names and the tea/maize framing increase learner engagement with the division problem? Note any learner comments that connected the mathematics to real land or food security concerns — these are evidence of the attitudes SLO being met and are worth sharing with the broader school community.

5. MODEL QUALITY — Were the cumulative models substantively different from a summary of today's notes, or did learners genuinely synthesise across lessons? If models looked like copied notes, the model-building phase instructions may need to be more explicitly comparative: 'Show me how today's lesson is connected to Lesson 3's decomposition technique — draw an arrow and write one sentence.'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners solved two composite polygon problems (an L-shaped school building footprint and a pentagonal community garden with a rectangular section removed) using both additive and subtractive area strategies, recording their working in a structured three-column evidence table and confirming that two different decompositions of the L-shape yield the same total area.
What did I learn?	Learners consolidated the principle that the area of any composite polygon equals the sum of its component standard-polygon areas, or a whole area minus a removed section — expressed as Area(composite) = Σ Area(components) ± Area(removed section) — and that this principle applies regardless of which valid decomposition strategy is chosen.
How does this explain the phenomenon?	Learners applied their full polygon-area toolkit (Lessons 1–7) to the anchoring phenomenon: they decomposed the Kericho seven-sided community plot into standard shapes, calculated the total area, divided it equally into three parts for the Wanjiku, Otieno, and Kosgei farming families, and produced a labelled diagram with a written justification of the fair division, preparing a complete solution for community presentation in Lesson 8.

LESSON 8 (80 minutes): Model Building, DQB Resolution & Final Assessment — Solving the Kericho Land Division Problem	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and assess learners' mastery of area calculations for all polygon types — triangles, quadrilaterals, heptagons, octagons, and irregular polygons — by applying the decomposition principle to resolve the Kericho community land-division problem.
Knowledge	Learners recall and integrate all formulae for area of triangles, rectangles, parallelograms, trapeziums, rhombuses, regular and irregular polygons, and the principle of decomposing any polygon into triangles and quadrilaterals.
Skills	Learners accurately decompose the seven-sided Kericho plot into familiar shapes, compute sub-areas, sum them to find total area, divide equally among three families, and defend their strategy with mathematical justification.
Attitudes	Learners appreciate that mathematics is a tool for social justice and equitable resource distribution, and reflect on the role of accurate measurement in protecting the livelihoods of Kenyan farming families.
Key Inquiry Question	How do we determine the area of polygons? How is the area of polygons applied in real-life situations?
Purpose in Storyline	This is the final lesson of the sub-strand. Groups present their full solutions to the Kericho land-division problem, defend their decomposition strategies, finalise the cumulative 'Area of Polygons Toolkit' poster, resolve all outstanding questions on the Driving Question Board, and complete an individual written assessment that spans all polygon types covered in lessons 1–8.
Safety Notes	No physical safety hazards. Remind learners to respect each other's presentations; critique the mathematics, not the person. Ensure equitable floor time — each group must have uninterrupted presentation time.

B. LESSON OVERVIEW
Lesson 8 is the capstone of Sub-Strand 2.5. It opens with a brief 'Predict' reflection where learners revisit their very first estimate of the Kericho plot area (made in Lesson 1) and compare it with their group's computed answer — surfacing how far their understanding has grown. Groups then formally present their land-division solutions, using their annotated polygon decomposition diagrams as evidence. A structured gallery walk and whole-class discussion serve as the 'Observe/Explain' phases, allowing learners to compare decomposition strategies, check for equivalent answers reached by different methods, and identify any errors. The class then finalises the cumulative 'Area of Polygons Toolkit' poster (the running model), annotating each polygon type with its formula, a Kenyan real-world example, and the decomposition principle. The Driving Question Board is revisited: every original question is answered in writing by the class, and any remaining wonderings (e.g., area of circles and sectors, Sub-Strand 2.6) are flagged. The lesson closes with a 25-minute individual written assessment covering all polygon types — triangles, quadrilaterals, regular polygons, and irregular polygons — all set in Kenyan contexts (tea farms, maize fields, rooftop solar panels, stadium floors in Nairobi).

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Overview of Ellipses,]	Each learner silently retrieves (or is shown) their Lesson 1 written prediction of the Kericho plot's total area. They write a one-sentence	Before distributing the prediction slips, say: 'Before we present our final solutions, I want you to look back at Day 1 — the very first time	Strategy: Metacognitive Comparison. Learners explicitly contrast prior intuitive thinking with evidence-based computation. This	Observable indicator: Every learner has a completed comparison sentence. Teacher scans for wild discrepancies (e.g.,

Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	reflection: 'My original estimate was ___ m². My group's computed answer is ___ m². The difference is ___ m² because ___.' Pairs share briefly. The class votes (hands up) on whether their estimate was too high, too low, or remarkably close, creating a quick data display on the board.	you saw this heptagonal plot in Kericho. What did you guess?' Distribute or display Lesson 1 prediction records. Allow 3 minutes of silent individual writing. Then say: 'Turn to your neighbour. Share your estimate versus your answer. You have 90 seconds.' After pairs share, conduct the hand-vote and record tallies on the board under 'TOO HIGH / TOO LOW / CLOSE'. Cold-call 3 learners: 'Amina, why do you think most of us overestimated?' (Wait 12 seconds.) 'Kipchoge, what changed in your thinking between Lesson 1 and today?' (Wait 10 seconds.) 'Wanjiku, what specific lesson helped you most?' (Wait 10 seconds.) Close by saying: 'Today we resolve the Kericho problem completely, defend our mathematics, and show the village elder — and ourselves — exactly how area of any polygon can be found.'	activates memory of the full learning arc, creates emotional investment in the resolution, and surfaces persistent misconceptions before the final assessment.	learner estimates still not close to the group answer) as a flag for targeted support during the gallery walk. The hand-vote tally informs pacing — if most learners were 'too high', briefly note that irregular shapes often trap dead space that our eyes overcount.
Observe Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group (of 4–5 learners) has 5 minutes to present their solution to the Kericho heptagonal plot problem. Presenters must: (1) show their annotated decomposition diagram on paper or the board, labelling every sub-shape and its dimensions; (2) state the formula used for each sub-shape; (3) give the total area in m²; (4) show how they divided the total area equally among the three farming families and mark the proposed dividing lines on the plot diagram. Observers (other groups) use a structured listening frame: 'I notice their decomposition used ___ triangles and ___ quadrilaterals. Their total area was ___ m². One question I have is ___.' After all presentations, a 5-	Set up presentation space before class: tape each group's poster-sized diagram to the wall or board. Introduce the session: 'You are now the mathematicians advising the Kericho village elder. Present your evidence clearly enough that a family's livelihood depends on it.' Use a visible 5-minute timer. After each group, ask the class: 'Does anyone get a meaningfully different total area? If so, let's examine why — different decompositions should give the same total.' (Wait 12 seconds. Cold-call one observer each time.) For the Gallery Walk, distribute sticky notes. After the walk, read out 2–3 sticky-note questions aloud: 'Group 3 asks Group 1: Why did you split the top portion	Strategy: Structured Argumentation and Evidence Comparison (NGSS Science and Engineering Practice 7 — Engaging in Argument from Evidence). Learners publicly defend their decomposition strategy and interrogate peers' approaches. Comparing multiple valid decompositions of the same polygon deepens understanding that area is an intrinsic property of the shape, not dependent on the method used to calculate it.	Observable indicators: (1) Each group's diagram correctly labels sub-shapes with dimensions and formulae. (2) Total areas across groups agree within a small rounding margin (acceptable ±0.5 m² given scale drawing). (3) Proposed dividing lines produce three sections each equal to one-third of total area. Teacher notes which groups could not achieve equal division — this signals a gap between computing area and applying it, which the written assessment will reveal individually.

	minute silent Gallery Walk allows learners to examine all posted diagrams and add one sticky-note question or affirming comment.	into two triangles rather than a trapezium? Group 1?' (Wait 12 seconds.) Validate equivalent approaches: 'Notice that Group 2 used five triangles and Group 4 used two triangles and one trapezium — both are correct decompositions, and they arrive at the same total area. That is a powerful mathematical truth.' Record all groups' total area answers on the board. If all are within rounding of each other, celebrate consensus: 'The village elder now has mathematical certainty.'		
Explain Phase 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class gathers around the running 'Area of Polygons Toolkit' poster that has been built incrementally across all 8 lessons. Each group is assigned one section of the poster to complete and annotate with: the polygon type name, its formula with labelled diagram, a Kenyan real-life example (e.g., 'Rhombus: a decorative tile floor in Jomo Kenyatta International Airport, Nairobi'), and a tick confirming 'decomposition works for this shape too'. Groups spend 8 minutes completing their assigned section. The class then does a 2-minute final review of the whole poster together, checking for completeness. A volunteer reads the decomposition principle statement aloud: 'Any polygon, no matter how irregular, can be decomposed into triangles and/or quadrilaterals whose areas can be found using known formulae. The total area of the polygon equals the sum of the areas of its parts.' The class signs the poster collectively.	Point to the poster: 'This toolkit has been built by all of us, lesson by lesson. Today we make it complete and final.' Assign sections quickly (pre-planned): Group 1 — triangles; Group 2 — quadrilaterals (rectangle, parallelogram, trapezium, rhombus); Group 3 — regular polygons (heptagon, octagon); Group 4 — irregular polygons and decomposition principle; Group 5 — Kenyan application examples row. Circulate during the 8 minutes. Prompt weak annotations: 'What is the exact formula? Write it with variables, not just words.' After the review, ask the full class: 'Is there any polygon type we have studied that is NOT on this toolkit?' (Wait 12 seconds.) Cold-call 2 learners. Then ask: 'Could you use this toolkit six months from now to solve a new land problem you have never seen before?' (Wait 10 seconds.) Invite one learner to read the decomposition principle aloud. Facilitate collective signing as a ritual of ownership.	Strategy: Collaborative Model Finalisation (NGSS Science and Engineering Practice 2 — Developing and Using Models). Physically completing and signing the cumulative toolkit transforms distributed lesson-by-lesson knowledge into an integrated, usable reference. Comparing Lesson 1's blank toolkit to today's completed version makes the learning arc visible and concrete.	Observable indicator: Every section of the poster contains the correct formula, a correctly labelled diagram, and a plausible Kenyan example. Teacher checks specifically that: (1) the decomposition principle is stated accurately, (2) irregular polygon section includes the Kericho heptagon as a worked example, (3) all formulae match KICD curriculum content. Any formula error is corrected publicly as a teaching moment before the written assessment.

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The DQB (posted since Lesson 1) is displayed prominently. Each original question card is read aloud by a learner volunteer. The class votes: 'CAN WE ANSWER THIS NOW?' For 'YES' cards, a learner writes a concise answer on the card in red marker. For 'PARTIALLY' cards, the partial answer is written and a note added: 'More in Sub-Strand 2.6.' For any 'NOT YET' cards (anticipated: questions about area of circles, curved boundaries), a new 'WONDER' card is created and posted on a 'Coming Soon — Sub-Strand 2.6' board. The driving question itself — 'How can we calculate the area of any polygon so that we can fairly divide the Kericho community land plot?' — is answered as a class in one final written statement that the teacher scribes on the board.	Stand beside the DQB: 'Eight lessons ago, you asked these questions because the Kericho plot confused you. Let's find out what we now know.' Read each card and invite a show-of-hands vote before calling on a learner to answer. For the driving question, say: 'Who would like to write our class answer on the board? It must be a complete sentence that a village elder with no mathematics background could understand.' (Wait 10 seconds. Invite a volunteer.) After the learner drafts the answer, ask: 'Class, does this answer fully explain how to divide the Kericho plot fairly? What would you add or change?' (Wait 12 seconds. Cold-call 2 learners.) Scribe the refined answer. Point to any 'WONDER' cards: 'These questions — about circles, sectors, curved farm boundaries — are exactly what Sub-Strand 2.6 will answer. You are already asking the right questions for next month's learning.' This creates anticipation and continuity.	Strategy: Driving Question Board Resolution (Anchoring Phenomena Closure). Publicly answering every original question from Lesson 1 creates a clear narrative of intellectual progress. It also models scientific practice — questions drive investigation, and investigations produce answers that generate new questions (NGSS Practice 1 — Asking Questions and Defining Problems).	Observable indicator: The class-constructed answer to the driving question correctly names (1) the decomposition principle, (2) the use of known formulae, and (3) the equal-division application to the Kericho plot. Teacher uses this as a whole-class formative check. Learners whose raised hands or verbal responses suggest incomplete understanding are noted for follow-up during marking of the individual written assessment.
Model Building Phase 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12	Learners complete a 25-minute individual written assessment. The paper contains 5 questions, each set in a Kenyan context, covering all polygon types from the sub-strand: Q1 — Find the area of a triangular maize plot near Eldoret with base 48 m and height 35 m (triangle formula). Q2 — A tea-drying platform in Kericho is a trapezium with parallel sides 22 m and 14 m, height 9 m; find its area (trapezium formula). Q3 — A community centre floor in Kisumu is a regular octagon with side 6 m and apothem 7.24 m; find its area (regular polygon formula). Q4 — An irregular plot in Nairobi's	Distribute papers face-down. Say: 'This is your individual assessment — it is your opportunity to show what you personally understand. You may look at the Toolkit poster at any time; it is a resource, not a cheat sheet. The maths must be yours.' (Flip papers.) Set a visible 25-minute timer. Circulate silently for the first 5 minutes to ensure all learners have started. Do not answer content questions; redirect with: 'Look at your Toolkit. Which formula applies here?' At the 10-minute mark, say quietly: 'You should be on Question 2 or 3 by now — manage your time.' At the 22-minute mark: 'Three minutes	Strategy: Authentic Performance Assessment in Kenyan Context (NGSS Practice 6 — Constructing Explanations; CBC Competency — Critical Thinking and Problem Solving). By situating every question in a Kenyan context — Eldoret maize plots, Kericho tea platforms, Kisumu community halls, Nairobi estates, Mombasa schools — the assessment reinforces that the mathematics of area is not abstract but is directly connected to land, livelihoods, food security, and infrastructure across Kenya. Allowing use of the Toolkit poster assesses application and reasoning rather than rote	Observable indicators for immediate scan after collection: (1) Q1–Q3: correct formula selection and arithmetic (benchmarks: Q1 = 840 m², Q2 = 162 m², Q3 ≈ 130.32 m²). (2) Q4: diagram is decomposed into at least 2 named sub-shapes; areas are summed; division is shown as total ÷ 2. (3) Q5: learner subtracts parallelogram area from rectangle area correctly. KICD rubric alignment — Exceeds Expectations: all five questions correct with clear working; Meets Expectations: Q1–Q3 and at least one of Q4–Q5 correct; Approaches Expectations:

🔍 Search ARES for similar readings	Eastlands estate has vertices that form a hexagon; a scale diagram is provided; decompose it into triangles and quadrilaterals, find total area, and suggest how it could be divided equally between two owners (decomposition principle, application). Q5 — A solar panel array on a school roof in Mombasa covers a region shaped like a parallelogram inside a larger rectangle; find the area of the non-solar region (composite shapes, subtraction). Learners work in silence. They may use the completed Toolkit poster on the wall.	remaining. Check your units and check that you have answered every part of each question.' Collect all papers at exactly 25 minutes. Immediately say: 'Well done. Before we finish, turn to the person next to you and tell them: one thing from this sub-strand you are proud of understanding.' (Allow 1 minute.) Close with: 'The three families in Kericho can now have their land divided fairly because of the mathematics you have learned in this classroom. That is real impact.'	formula recall.	Q1–Q2 correct, attempts Q3–Q5; Below Expectations: fewer than two questions fully correct. Teacher uses results to plan Sub-Strand 2.6 entry points and identify learners needing remediation.

D. TEACHER REFLECTION

After Lesson 8, reflect on the following: (1) Assessment evidence — Did the individual written results reveal mastery gaps in any specific polygon type (e.g., regular polygons, composite shapes)? Which learners are at 'Below Expectations' and what targeted support will I offer before Sub-Strand 2.6? (2) Group presentations — Did all groups successfully divide the Kericho plot into three equal areas, or did some stop at computing total area? If the equal-division step was weak, plan a bridging activity linking area to fair-sharing fractions. (3) DQB resolution — Were there any questions on the DQB that the class could not fully answer? Document these and revisit them explicitly in Sub-Strand 2.6 (area of a part of a circle). (4) Toolkit quality — Is the completed 'Area of Polygons Toolkit' poster accurate enough to remain on the classroom wall as a reference for Sub-Strand 2.6? Check all formulae for correctness before leaving the poster up. (5) Phenomenon closure — Did learners explicitly connect their mathematics to the Kericho families' real situation? If the human/social justice dimension felt thin, plan a brief warm-up for Sub-Strand 2.6 that returns to the Kericho context and asks: 'What if part of the plot boundary was curved — how would that change our calculation?' This sustains the phenomenon thread across sub-strands.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Groups presented their annotated decomposition diagrams of the seven-sided Kericho heptagonal plot, showing how they split it into triangles and quadrilaterals, computed sub-areas using known formulae, and proposed boundary lines dividing the total area equally among three farming families. Multiple valid decomposition strategies were compared and shown to produce equivalent total areas.
What did I learn?	Any polygon — no matter how many sides or how irregular — can be decomposed into triangles and quadrilaterals whose areas can be calculated using known formulae. The total area equals the sum of the parts. This principle, together with the formulae for all polygon types, gives us a complete toolkit to solve any land-area problem in Kenya or beyond.
How does this explain the phenomenon?	We resolved the driving question: the Kericho community land plot has a calculable total area found by decomposition, and it can be divided fairly among three families by allocating boundary lines so that each section equals exactly one-third of the total area. The 'Area of Polygons Toolkit' poster and DQB resolution show that every question raised in Lesson 1 has now been answered

	through eight lessons of investigation.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.